



Accredited with Grade A+ by NAAC, GoI

ACADEMIC REGULATIONS & SYLLABI



Faculty of Management Studies
Bachelor of Business Administration (BBA)
Business Analytics
Batch: 2024-27



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY - CHARUSAT

Accredited with Grade A+ by NAAC, GoI

Education Campus – Changa, (ECC), hitherto a conglomerate of institutes of professional education in Engineering, Pharmacy, Computer Applications, Management, Applied Sciences, Physiotherapy and Nursing, is one of the choicest destinations by students. It has been transformed into Charotar University of Science and Technology (CHARUSAT) through an Act by Government of Gujarat. CHARUSAT is permitted to grant degrees under Section-22 of UGC – Government of India.

The journey of CHARUSAT started in the year 2000, with only 240 Students, 4 Programmes, one Institute and an investment of about Rs. 3 Crores (INR 30 million). At present there are nine different institutes falling under ambit of six different faculties. The programmes offered by these faculties range from undergraduate (UG) to Ph.D degrees. These faculties, in all offer 74 different programmes. A quick glimpse is as under:

Faculty	Institute	Programmes Offered
Faculty of Technology and Engineering	Chandubhai S. Patel Institute of Technology	B.Tech M.Tech Ph.D
Faculty of Pharmacy	Ramanbhai Patel College of Pharmacy	B.Pharm M.Pharm Ph.D PGDCT / PGDPT
Faculty of Management Studies	Indukaka Ipcowala Institute of Management	BBA MBA Ph.D
Faculty of Computer Applications	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications.	BCA MCA M.Sc IT Ph.D BCA
Faculty of Sciences	P.D. Patel Institute of Applied Sciences	B.Sc M.Sc Ph.D
Faculty of Medical Sciences	Ashok and Rita Institute of Physiotherapy	B.PT M.PT Ph.D

	Manikaka Topawala Institute of Nursing	B.Sc (Nursing) P.B. B.Sc. Nursing GNM M.Sc. Ph.D
	Bapubhai Desaibhai Patel Institute of Paramedical Sciences	BMIT BSMT BOTAT BOP MSc. MIT MSMLT MSOTAT PGDMLT PGDCH Ph. D

The development and growth of the institutes have already led to an investment of over Rs.150 Crores (INR 1500 Million). The future outlay is planned with an estimate of Rs.250 Crores (INR 2500 Million).

The University is characterized by state-of-the-art infrastructural facilities, innovative teaching methods and highly learned faculty members. The University Campus sprawls over 125 acres of land and is Wi-Fi enabled. It is also recognized as the Greenest Campus of Gujarat.

CHARUSAT is privileged to have 360 core faculty members, educated and trained in IITs, IIMs and leading Indian Universities, and with long exposure to industry. It is also proud of its past students who are employed in prestigious national and multinational corporations.

From one college to the level of a forward-looking University, CHARUSAT has the vision of entering the club of premier Universities initially in the country and then globally. **High Moral Values like Honesty, Integrity and Transparency** which has been the foundation of ECC continues to anchor the functioning of CHARUSAT. Banking on the world class infrastructure and highly qualified and competent faculty, the University is expected to be catapulted into top 20 Universities in the coming five years. In order to align with the global requirements, the University has collaborated with internationally reputed organizations like Pennsylvania State University – USA, University at Alabama at Birmingham – USA, Northwick Park Institute –UK, ISRO, BARC, etc.

CHARUSAT has designed curricula for all its programmes in line with the current international practices and emerging requirements. Industrial Visits, Study Tours, Expert Lectures and Interactive IT enabled Teaching Practice form an integral part of the unique CHARUSAT pedagogy.

The programmes are credit-based and have continuous evaluation as an important feature. The pedagogy is student-centered, augurs well for self-learning and motivation for enquiry and research, and contains innumerable unique features like:

- Participatory and interactive discussion-based classes.
- Sessions by visiting faculty members drawn from leading academic institutions and industry.
- Regular weekly seminars.
- Distinguished lecture series.
- Practical, field-based projects and assignments.
- Summer training in leading organizations under faculty supervision in relevant programmes.
- Industrial tours and visits.
- Extensive use of technology for learning.
- Final Placement through campus interviews.

Exploration in the field of knowledge through research and development and comprehensive industrial linkages will be a hallmark of the University, which will mould the students for global assignments through technology-based knowledge and critical skills.

The evaluation of the student is based on grading system. A student has to pursue his/her programme with diligence for scoring a good Cumulative Grade Point Average (CGPA) and for succeeding in the chosen profession and life.

CHARUSAT welcomes you for a Bright Future.....



CHAROTAR UNIVERSITY OF SCIENCE AND
TECHNOLOGY

Faculty of Management Studies

Accredited with Grade A+ by NAAC, GoI

ACADEMIC REGULATIONS

BBA Programme

Bachelor of Business Administration (BBA) Programme

Business Analytics

Academic Year 2024-25

Charotar University of Science and Technology (CHARUSAT)

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CHARUSAT

FACULTY OF MANAGEMENT STUDIES

ACADEMIC REGULATIONS

Bachelor of Business Administration (BBA)

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a specified load of course work in the chosen subject of specialization and also complete a project/dissertation if any.

2. Duration of Programme

2.1 The Bachelor of Business Administration (BBA) – *Business Analytics* programme of Charotar University of Science and Technology (CHARUSAT) is a **three year full-time** under-graduate programme, leading to the award of the degree of Bachelor of Business Administration (BBA).

2.2 The span of the programme is six years (as decided by the university from time to time) from the date of registration in the programme.

3. Eligibility and mode of admissions

Candidate should have passed 12th standard (or equivalent) examination (any stream) with English as a subject is eligible for admission to the BBA – Business Analytics Programme and according to the regulations for admission decided by CHARUSAT from time to time.

4. Programme structure and Credits

4.1 *A student admitted to a program should study the course and earn credits specified in the course structure.* The details of programme structure, credit requirements, areas of specialisation proposed to be offered, etc. are presented at Appendix – I.

4.2 Programme Outcomes (PO)

Graduates will be able to...

PO1 - Analyse contemporary issues and apply management knowledge for resolution.

PO2 - Practice effective communication and interpersonal skills.

PO3 - Understand Global Perspectives and exhibit Cross-Cultural sensitivity.

PO4 - Demonstrate Attitude for continuous learning and development.

5. Attendance

5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will require to be condoned by the Dean/Principal.

5.2 Student attendance in every course should be 80%.

6. Course Evaluation

6.1 The performance of every student in each course will be evaluated as follows:

- 6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 50% of the marks for the course; and
- 6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these, for 50% of the marks for the course.

6.2 University Examination

- 6.2.1 The final examination by the University for 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation or a combination of these.
- 6.2.2 In order to earn the credit in a course, a student has to obtain a grade other than F (Fail) or Ab (Absent)

6.3 Performance at Internal Evaluation Components and University Examination

If a student secures minimum passing marks of 36% in the University examinations in any course but fails to obtain the minimum passing total percentage of 36%, he/she has to repeat the university examination in the course.

Minimum percentage marks in University Exam for pass in any course	Minimum total percentage marks (i.e. internal+ University) for pass in any course
36%	36%

7. Grading

7.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Grading Scheme:

Range of Marks (%)	≥96	≥ 86 and <95	≥ 76 and <86	≥ 66 and <76	≥ 56 and <66	≥ 46 and <56	≥ 36 and <46	<35
Grade	O	A+	A	B+	B	C	P	F
Grade Point	10	9	8	7	6	5	4	0

7.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move to the next academic year if he/she fails to clear all the courses at the end of the respective academic year.

(iv) in addition to above, a student has to comply with the requirements of the regulatory bodies, wherever such requirements exist.

*(*Whenever the university conducts the examinations of that course, it will be considered as a chance, irrespective of whether the student appears for the examination or not.)*

8. Awards of Bachelor Degree

8.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

8.1.1 He should have earned at least minimum required credits as prescribed in course structure.

8.2 Any student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree.

9. Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

First Class with Distinction	CGPA $\geq$ 7.0
First Class	CGPA $\geq$ 6.0
Second Class	CGPA $\geq$ 5.0
Pass Class	CGPA $\geq$ 3.6

10. Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA and class obtained.

Appendix – I

Details of BBA – Business Analytics Programme Structure, Credit Requirements and Specialisation

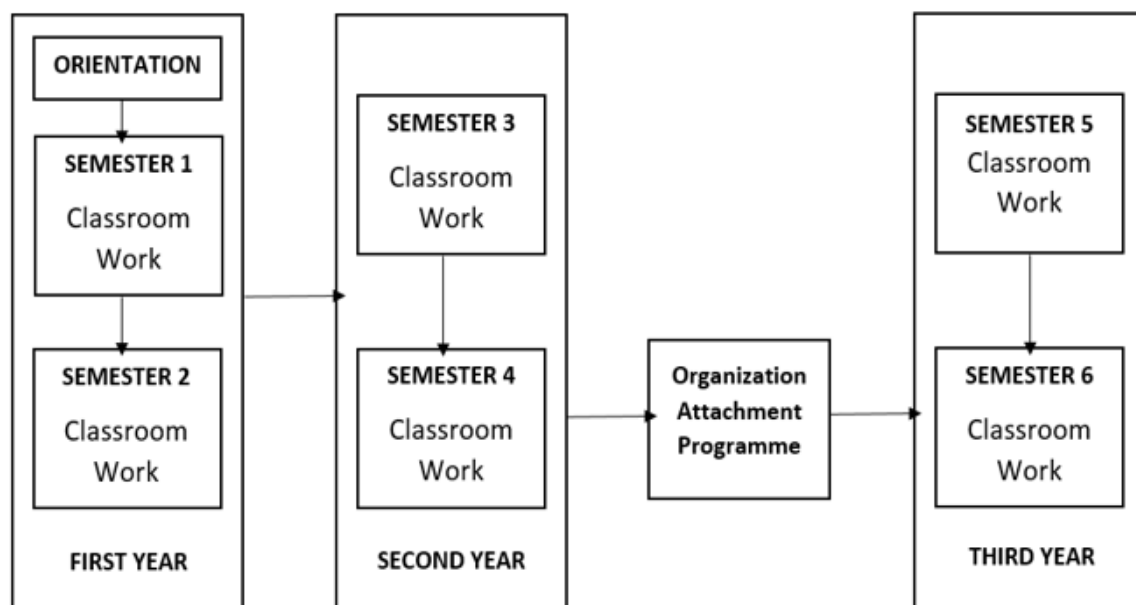
A1. Programme Structure

A1.1 The programme is structured into *six semesters*, consisting of classroom contact-based work and *Organization Attachment/Internship Programme*.

A1.2 Each semester will be for a minimum of *90 working days* for classroom work, covering classroom contact sessions, laboratory/tutorial/library/group work, case discussions and presentation, field-based as well as library/internet search-based assignments and projects, classroom exercises, management and simulation games, short quizzes, and class tests. The duration for any organizational attachment/training during the semester and final University examinations will be in addition to the 90 working days.

A1.4 The structure of the BBA programme is as shown in the following figure:

Figure A1: BBA – Business Analytics Programme Structure



A2. Credits

A2.1 Any student of the BBA – Business Analytics programme who earns 140 credits by pursuing the prescribed course work and passing all tests, examinations, assignments, laboratory work, projects and all other evaluation components as per the passing standards of the University will be eligible for the award of the Bachelor of Business Administration (BBA).

- A2.2 The current distribution of credits over the Three-year period for classroom contact sessions and laboratory/ tutorial/ library/ group work sessions will be as follows:

TableA2: Semester-wise Distribution of Credits

Sl. No.	Semester	Number of Credits
1	Semester - 1	22
2	Semester - 2	22
3	Semester - 3	24
4	Semester - 4	24
5	Semester - 5 (Organization Attachment Programme) (18+4)	22
6	Semester – 6	20
	Total Credits	134

- A2.3 Some courses will have only classroom contact sessions and some others will have tutorial/ laboratory/ library/ group work sessions, as shown in the list of courses.

A3. *Courses, Curricula and Revision*

- A3.1 The Faculty Board of the Faculty of Management Studies and the Dean of the Faculty of Management Studies will keep the curricula current and in tune with the changes happening in the world of management and make it relevant to the needs of different organs of society.
- A3.2 The review of the programme, its structure, the course curricula, pedagogy and evaluation will be undertaken by the Boards of Studies.

Bachelor of Business Administration – Business Analytics

Sem.	Course Code	Name of the Course	Nature	Credits	Remarks
1	BBU101	Principles and Practices of Management	Theory	04	-
	BBU102	Logical Analysis and Decision Making	Theory	04	-
	BBU103	Computer Fundamentals and Applications	Practical	04	-
	BBU104	Skills for Management Professionals	Practical	04	-
	HSUA101	Communicative English	Practical	02	Offered by FOH
	CLUV101	Environmental Sciences	Practical	02	Offered by FTE
	CUUV101	Community Engagement and Sustainable Development	Practical	02	Students can choose One VAC
	CUUV103	Physical Education and Sports	Practical	02	
Total Credit for the Semester:				22	
2	BBU151	Fundamentals of Economics	Theory	04	-
	BBU152	Introduction to Data Science	Theory	04	-
	BBU153	Basics of Quantitative Methods	Theory	04	-
	BBU154	Introduction to Computing	Practical	04	-
	BBU155	Business Case Analysis	Theory	04	-
	HSUSI01 – HSUSI17	A Course on Liberal Arts	Practical	02	Offered by FOH
Total Credit for the Semester:				22	
Total Credits for First Year				44	
3	BBU201	Financial Management	Theory	04	-
	BBU202	Human Resources Management	Theory	04	-
	BBU203	Operations Research Models	Theory	04	-
	BBU204	Database Management Systems	Practical	04	-
	BBU205	Marketing Management	Theory	04	-
	HSI21.02 D	Creativity, Problem Solving & Innovations	Practical	02	Offered by FOH
	BMUD201	Banking & Insurance (UE)	Practical	02	University Elective
Total Credit for the Semester:				24	

Sem.	Course Code	Name of the Course	Nature	Credits	Remarks
4	BBU251	Fundamentals of Research	Theory	04	-
	BBU252	Advanced Quantitative Methods	Practical	04	-
	BBU253	Entrepreneurship Development and Innovation Management	Practical	04	-
	BBU254	Predictive Business Analytics using R	Practical	04	
	BBU255	Data Visualization and Communication	Practical	04	
	HSIII.03 D	Human Values & Professional Ethics	Practical	02	<i>Offered by FOH</i>
	BMUD251	Healthcare Management (UE)	Practical	02	<i>University Elective</i>
	CUUV102	National Cadet Corps	Practical	02	<i>VAC for selected students</i>
Total Credit for the Semester:				24	
Total Credits for Second Year				48	
5	BBU301	Organizational Attachment Program	Practical	04	-
	BBU302	Strategic Management	Theory	04	-
	BBU303	Supply Chain & Logistic Management	Theory	04	-
	BBU304	Machine Learning	Practical	04	-
	BBU305	Analytics for Marketing	Practical	04	-
	HSUA301	Communication and Soft Skills	Practical	02	<i>Offered by FOH</i>
Total Credit for the Semester:				22	
6	BBU351	Social Media & Web Analytics	Practical	04	-
	BBU352	Grand Project	Practical	06	-
	BBU353	Business Forecasting for Time-Series	Practical	04	-
	BBU354	Analytics for HR	Practical	04	-
	HS132.03 D	Contributory Personality Development	Practical	02	<i>Offered by FOH</i>
Total Credit for the Semester:				20	
Total Credits for Third Year				42	
Total Credits of 3 Years' BBA Degree Programme				134	

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-I

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BBU101	Principles and Practices of Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU102	Logical Analysis and Decision Making	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU103	Computer Fundamentals and Applications	04	---	4.00	---	4.00	50	50	100	---	---	---
BBU104	Skills for Management Professionals	04	---	4.00	---	4.00	50	50	100	---	---	---
HSUA101	Communicative English	02	---	2.00	---	2.00	---	---	---	25	25	50
CLUV101	Environmental Sciences	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	20			---	20.00			400			100
Elective Group-1												
CUUV101	Community Engagement and Sustainable Development	02	---	2.00	---	3.00	---	---	---	---	---	---
CUUV103	Physical Education and Sports	02	---	2.00	---	3.00	---	---	---	---	---	---

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-II

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BBU151	Fundamentals of Economics	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU152	Introduction to Data Science	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU153	Basics of Quantitative Methods	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU154	Introduction to Computing	04	---	4.00	---	4.00	50	50	100	---	---	---
BBU155	Business Case Analysis	04	4.00	---	---	4.00	50	50	100	---	---	---
HSUS101 – HSUS117	A Course on Liberal Arts	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	22				22.00			500			50

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-III

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BBU201	Financial Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU202	Human Resources Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU203	Operations Research Models	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU204	Database Management Systems	04	---	4.00	---	4.00	50	50	100	---	---	---
BBU205	Marketing Management	04	4.00	---	---	4.00	50	50	100	---	---	---
HS12L02 D	Creativity, Problem Solving & Innovations	02	---	2.00	---	2.00	---	---	---	25	25	50
BMUD201	Banking & Insurance (UE)	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	24				24.00			500			100

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-IV

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BBU251	Fundamentals of Research	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU252	Advanced Quantitative Methods	04	---	4.00	---	4.00	50	50	100	---	---	---
BBU253	Entrepreneurship Development and Innovation Management	04	---	4.00	---	4.00	50	50	100	---	---	---
BBU254	Predictive Business Analytics using R	04	---	4.00	---	4.00	50	50	100	---	---	---
BBU255	Data Visualization and Communication	04	---	4.00	---	4.00	50	50	100	---	---	---
HS111.03 D	Human Values & Professional Ethics	04	---	4.00	---	4.00	50	50	100	---	---	---
BMUD251	Healthcare Management (UE)	02	---	2.00	---	2.00	---	---	---	25	25	50
CHUV102	National Cadet Corps	02	---	2.00	---	3.00	---	---	---	---	---	---
	Total	24				24.00			600			50

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-V

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BBU301	Organizational Attachment Program	04	---	4.00	---	4.00	---	---	---	50	50	100
BBU302	Strategic Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU303	Supply Chain & Logistic Management	04	4.00	---	---	4.00	50	50	100	---	---	---
BBU304	Machine Learning	04	---	4.00	---	4.00	---	---	---	50	50	100
BBU305	Analytics for Marketing	02	---	2.00	---	2.00	---	---	---	25	25	50
HSUA301	Communication and Soft Skills	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	20				20.00			200			300

ANNEXURE – A – BBA (Business Analytics) TEACHING & EXAMINATION SCHEME
BBA-BA – SEMESTER-VI

Course Code	Course Name	Credits	Teaching Scheme/Contact Hours				Evaluation Scheme					
			Theory	Practical	Contact Hours	Total Hours	Theory			Practical		
							Internal	External	Total	Internal	External	Total
BBU351	Social Media & Web Analytics	04	---	4.00	---	4.00	---	---	---	50	50	100
BBU352	Grand Project	06	---	6.00	---	4.00	---	---	---	50	50	100
BBU353	Business Forecasting for Time-Series	04	---	4.00	---	4.00	---	---	---	50	50	100
BBU354	Analytics for HR	04	---	4.00	---	4.00	---	---	---	50	50	100
HS132.03 D	Contributory Personality Development	02	---	2.00	---	2.00	---	---	---	25	25	50
	Total	20				20.00						450

Bachelor of Business Administration (BBA) – Business Analytics Programme

SYLLABI

(Semester – I)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

BBU101: PRINCIPLES AND PRACTICES OF MANAGEMENT
YEAR 1, SEMESTER – I

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Management is crucial for individuals as it equips them with the skills and knowledge necessary to effectively lead and navigate complex organizational structures, fostering success in diverse professional environment. The multifaceted nature of management equips students with a holistic understanding of key management principles.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Management	10
2	Planning and Forecasting	15
3	Organizing	15
4	Staffing	10
5	Directing and controlling	10
Total Contact Hours		60

Course Syllabus

Module 1: Introduction to Management **10 hours**

A Conceptual Framework and Application of management, Development of Management Thought, Process of Management, Levels of Management, Roles of an Effective Manager

Module 2: Planning and Forecasting **15 hours**

Planning process, Planning premises, Types of plans – based on breadth and use, Techniques of forecasting, Business barometers, Significance and limitations of forecasting

Module 3: Organizing **15 hours**

Principles of organizing , Factors affecting organizational Structure , Departmentation and Delegation , Span on Management, Forms of Organizational Structure

Module 4: Staffing **10 hours**

Human Resource Planning, Job Analysis and Job Description, Job Specification, Recruitment, Selection, Training and Development, Placement and Orientation

Module 5: Directing and controlling **10 hours**

Principles of Directing, Directing and Human Factor, Theories of leadership, Theories of leadership, Principles of effective control system, Techniques of control

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 2 hours based on written paper for 50 marks.

Text Book:

1. L. M. Prasad, *Principles and Practice of Management, Latest Edition*, Sultan Chand and Sons.

Reference Books:

1. Karminder Ghuman and K Aswathappa, *Management: Concept, Practice and Cases*, Latest Edition, Mc Graw Hill Education.
2. T. Ramasamy, *Principles of Management*, Latest Edition, Himalaya Publishing House.
3. Stoner, Freeman and Gilbert Jr, *Management*, Latest Edition, Prentice Hall of India

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the nature, role and significance of management
CO2	Analyze the functions of management
CO3	Develop and evaluate the recruitment and selection process
CO4	Execute the leadership practices and apply the suitable techniques of controlling

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction to Management	✓			
2	Planning and Forecasting		✓		
3	Organizing		✓		
4	Staffing			✓	
5	Directing and controlling				✓

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understand</i> the nature, role and significance of management	2	2	2	3
CO2	<i>Analyze</i> the functions of management	3	3	2	3
CO3	<i>Develop</i> and evaluate the recruitment and selection process	2	3	2	2
CO4	<i>Execute</i> the leadership practices and apply the suitable techniques of controlling	3	2	2	3

BBU102: LOGICAL ANALYSIS AND DECISION MAKING
YEAR 1, SEMESTER – I

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Decision Making is one of the vital skill which a manager come across while handling the routine tasks. Along with the decision making, a Manager is expected to have certain abilities which involves critical and logical way of thinking. This course Logical Analysis and Decision Making will accustom students to analyze day-to-day tasks and will develop students as managers with independent, self-directed thinking.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline:

Module Number	Title of the Module	Contact Hours
1	Logic and Reasoning	10
2	Deductive Patterns of Reasoning	05
3	Inductive Pattern of Reasoning	05
4	Informal Logic	15
5	Logical Fallacies	10
6	Researching and Writing Arguments	05
Total Contact Hours		60

Course Syllabus

Module 1: Logic and Reasoning

10 hours

Introduction to basic concepts of logic, Statement Vs Sentence, Statement: True, False, and Controversial, Identification of Premise Indicators, Identification of Conclusion Indicators, Identification of Conclusion Indicators, Concept of an Argument

Module 2: Deductive Patterns of Reasoning

05 hours

Concept of Deductive Argument, Hypothetical Syllogism, Categorical Syllogism, Argument by Elimination, Argument Based on Mathematics, Argument from definition

Module 3: Inductive Pattern of Reasoning

05 hours

Concept of Inductive argument, Inductive Generalization, Predictive Argument, Argument from Authority, Causal Argument, Statistical Argument

Module 4: Informal Logic

15 hours

Categorical Logic, Propositional Logic

Module 5: Logical Fallacies

10 hours

Fallacies of Relevance, Fallacies of insufficient Evidence

Module 6: Researching and Writing Arguments

15 hours

Identifying and Justifying Statements, Non-Statements, Premises, Conclusions, and arguments, Identifying the Pattern of Reasoning Involved in the Arguments, Identifying Fallacies Present in the Arguments, Thinking Critically about Media, Thinking Critically about Media

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 2 hours based on written paper for 50 marks.

Text Book:

1. Irving M. Copi and Carl Cohen Come, Introduction to Logic, Latest Edition, Pearson.

Reference Books:

1. Harry J. Gensler, Introduction to Logic, Routledge Publisher, Latest Edition, Routledge Publisher.
2. James B. Nance and Douglas J. Wilson, Introductory Logic: Student, Canon Press, Latest Edition.
3. Nripendra Nath Biswas, Introduction to logic and switching theory, Gordon and Breach Science Publishers, Latest Edition.

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Develop logical reasoning and decision-making capabilities.
CO2	Analyze reason and present a valid argument.
CO3	Evaluate Critically day-to-day tasks and make sound decision making.
CO4	Understand the basics of formal logic.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Logic and Reasoning				√
2	Deductive Patterns of Reasoning	√			
3	Inductive Pattern of Reasoning	√			
4	Informal Logic		√		
5	Logical Fallacies			√	
6	Researching and Writing Arguments	√			

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	Develop logical reasoning and decision-making capabilities.	2	2	2	3
CO2	Analyze reason and present a valid argument.	3	3	2	3
CO3	Evaluate Critically day-to-day tasks and make sound decision making.	2	3	2	2
CO4	Understand the basics of formal logic.	3	2	2	3

BBU103: COMPUTER FUNDAMENTAL AND APPLICATIONS
YEAR 1, SEMESTER – I

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

In today's world, computers have become a basic necessity of both personal and professional life. In the business world, from small-scale businesses to large corporations, computers are used for every task. A person without the knowledge of computers is considered illiterate in a way, as it would be impossible to progress without computer skills. This course is aimed at giving the students a basic understanding of computers and programs that are used in management, day in and day out.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction Computer Applications & Working with MS Word	10
2	Working with MS PowerPoints	12
3	Working with MS Excel: Essentials	10
4	Working with MS Excel: Formulas	14
5	Working with MS Excel: Charts and Tables	14
Total Contact Hours		60

Course Syllabus

Module 1: Introduction Computer Applications & working with MS word **10 hours**

Types of peripherals, Networking and Internet, Types of software, Introduction - MS Word, Using templates, Formatting, Enhancing word documents, Working with tables, columns, and sections, Managing mailing and forms, Creating special documents

Module 2: Working with MS PowerPoint **12 hours**

Introduction, Essential features, Advanced presentation formatting, themes, and masters, Clip art, pictures, smart art, Animation, transition, multimedia, Delivering presentation and creating support material

Module 3: Working with MS Excel: Essentials **10 hours**

Navigating worksheets, Editing, copying, moving, printing, Layout setting, Themes management, Manipulation, Conditional formatting, Naming range, Format painter, Working with columns and rows, Referencing: relative / absolute

Module 4: Working with MS Excel: Formulas **14 hours**

Working with formula, Financial functions, Logical functions, Statistical functions, Date-time functions, Text functions, Other functions

Module 5: Working with MS Excel: Charts and Tables **14 hours**

Introduction, Creating charts, Modifying charts, Chart elements, Combination charts, Excel tables, Pivot table

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Text Book:

1. Joe Habraken, Inside OUT: Microsoft Office 2019, Latest Edition, Pearson.

Reference Books:

1. Greg Harvey, Microsoft Excel 2019 All in One for Dummies, Latest Edition, Wiley.
2. Theodor Richardson and Charles Thies, Microsoft Office 2013/365 and Beyond: Computer Concepts and Applications, Latest Edition, Mercury Learning and Information.

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the concept of input and output devices of computer.
CO2	Develop working skills with the knowledge of computer software.
CO3	Create basic word documents, spreadsheets and presentations.
CO4	Apply computer skills to their assignments and later in work life.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction Computer Applications & Working with MS Word	√			
2	Working with MS PowerPoints		√		
3	Working with MS Excel: Essentials			√	
4	Working with MS Excel: Formulas				√
5	Working with MS Excel: Charts and Tables				√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understand</i> the concept of input and output devices of computer.	3	2	2	2
CO2	<i>Develop</i> working skills with the knowledge of computer software.	3	2	1	1
CO3	<i>Create</i> basic word documents, spreadsheets and presentations.	3	2	2	1
CO4	<i>Apply</i> computer skills to their assignments and later in work life.	3	2	1	1

BBU104: SKILLS FOR MANAGEMENT PROFESSIONALS
YEAR 1, SEMESTER – 2

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	02	30

Rationale

This course will explore traditional and contemporary challenges, practices, and/or events faced by individuals, teams, and leaders in today's organizations through in-class discussion. Clear, precise, and effective communication has become a sine qua non in today's information-driven world given its interdependencies and seamless connectivity. Any aspiring professional cannot but master the key elements of such communication. The objective of this course is to equip students with the necessary management skills to comprehend and successfully convey any idea, technical or otherwise, as well as give them the necessary polish to become persuasive communicators.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Connect over Communicate	05
2	Connecting via Storytelling	05
3	Personal Communication Skills	05
4	Striking Strategic Conversations	05
5	Effective Management	05
6	Contemporary Issues	05
Total Contact Hours		30

Course Syllabus

Module 1: Connect over Communicate

05 hours

Influence Matters – The What, Why, and How? What is influence? How to set influence in motion? Rhetoric and Persuasion, the potential of persuasion, “Powering” Persuasion, Persuasion Triggers, why create influence? Discuss cases of persuasive managerial communication.

Module 2: Connecting via Storytelling

05 hours

Why stories work to connect? The Story Matrix, The IGNITE approach to Storytelling, Types of Stories, telling intelligent stories, Types of intelligences, Fit your story to your type of intelligence

Module 3: Personal Communication Skills

05 hours

Your Inner Brand, Communication Quotient, Your Interpersonal Skills, Networking.

Module 4: Striking Strategic Conversations

05 hours

Need for Strategic Dialogue, Technologically Mediated Strategic Dialogue, Presentations for Effective Strategic Dialogue, Key Tips for making Effective Presentations

Module 5: Effective Management

05 hours

Fundamentals of Management Learning and Negotiation, Critical Elements of Effective Management Learning and Negotiation, Reflexivity, Context, Critical Thinking Strategies of Effective Management Learning and Negotiation, Managerial Writing, Managerial Listening, Verbal Communication, on-Verbal Communication.

Module 6: Contemporary Issues

05 hours

Contemporary Issues

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Text Book:

1. Geraldine E. Hynes, Managerial Communication (4th Edition), Tata McGraw-Hill

Reference Books:

1. Andy Green, CIPR, Effective Personal Communication Skills for Public Relations, latest Edition, Pearson Education,
2. Charles Fleetham, T, The Search for Unrational Leadership, Latest Edition, MacMillan India
3. John Hartley; Routledge, Communication, Cultural, and Media Studies, The Key Concepts, Latest Edition, Pearson Education,
4. Russel H. Granger, The 7 Triggers to Yes (Latest Edition), Latest Edition, Tata McGraw-Hill
5. Maris Roze & Simon Maxwell; Technical Communication in the Age of the Internet (Fourth Edition, Pearson Education

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Use effective managerial communication to create connections through influence
CO2	Understand the key roles of story-telling in persuasive communication
CO3	Interpret the strategic development of personal communication skills
CO4	Use effective presentation skills for strategic dialogues
CO5	Develop critical thinking skills for effective management and negotiation
CO6	Evaluate the fore issues and challenges encountered in developing managerial skills in contemporary times

Course Outcomes Mapping

Module	Module Name	Course Outcomes					
		CO1	CO2	CO3	CO4	CO5	CO6
1	Connect over Communicate	✓					
2	Connecting via Storytelling		✓				
3	Personal Communication Skills			✓			
4	Striking Strategic Conversations				✓		
5	Effective Management					✓	
6	Contemporary Issues						✓

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution	<i>Practice</i> effective communication and interpersonal skills	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity	<i>Demonstrate</i> Attitude for continuous learning and development
CO1	<i>Use</i> effective managerial communication to create connections through influence	1	1	3	2
CO2	<i>Understand</i> the key roles of story-telling in persuasive communication	2	1	3	2
CO3	<i>Interpret</i> the strategic development of personal communication skills	2	2	3	3
CO4	<i>Use</i> effective presentation skills for strategic dialogues	1	1	3	1
CO5	<i>Develop</i> critical thinking skills for effective management and negotiation	2	2	3	3
CO6	<i>Evaluate</i> the fore issues and challenges encountered in developing managerial skills in contemporary times	2	2	3	2

Charotar University of Science and Technology(CHARUSAT)

Faculty of Humanities (FoH) Semester- I HSUA101: Communicative English

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	02	2
Marks	--	50	--	50	

Pre-requisite:

- Basic knowledge of English Language & Communication

Objectives of the Course:

- To make students aware of and give them first-hand experience of Communicative English
- To make the students aware of the different communicative functions of English and help them communicate using those functions
- To assist and guide the learners to be able to listen to and to read a text and identify specific and global information
- To assist and guide the learners to be able to speak and to write clearly and effectively

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Communicative English	03
2.	Communication Functions and Grammar	06
3.	Communication Skills I – Listening	03
4.	Communication Skills II – Reading	03
5.	Communication Skills III – Speaking	06
6.	Communication Skills IV – Writing Composition	06
7.	Developing Vocabulary	03
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	Introduction to Communicative English	03 Hours	10%
	English as a Window Language; Varieties of English: British English, American English, Indian English; Language Variations; Importance of English for Academic and Professional Development; Strategies for Language Acquisition; Formal VS Informal English		
2.	Communication Functions and Grammar	6 Hours	20%
	Communicative Grammar: Usage of parts of speech in communicative way, Sentence Structure & Tense, Active Passive & Direct-Indirect Greeting and Introducing; Making Requests and Asking for Information; Expressing Likes and Dislikes; Seeking Permission; Giving and Taking Advice; Describing People, Place, Things; Retelling Past Events; Comparing and Contrasting; Persuading; Describing Cause and Effect		
3.	Communication Skills I – Listening	3 Hours	10%
	Importance of Listening as a Language Skill; Basic Listening Skills; Types of Listening; Barriers to Listening; Strategies for Effective Listening; Listening Practice		
4.	Communication Skills II – Reading	3 Hours	10%
	Importance of Reading as a Language Skill, Reading Strategies: Skimming, Scanning, Intensive Reading, Extensive Reading, Strategies for Effective Reading Comprehension, Reading Practice		
5.	Communication Skills III – Speaking	6 Hours	20%
	Importance of Speaking as a Language Skill, Basic Speaking Skills; Paralanguage for Effective Speaking; Strategies for Oral Communication; Extempore and Public Speaking & Presentation		
6.	Communication Skills IV – Writing Composition	6 Hours	20%
	Importance of Writing as a Language Skill; Process of writing: Prewriting, Drafting, Revision, Editing, Publication; Seven C's of Writing; Sentence Construction – Complex, Compound, Paragraph Development; Letter Writing (Academic Context)		
7.	Developing Vocabulary	3 Hours	10%
	High Frequency Vocabulary (Every day and Academic use); Words Often Confused and Misused; Useful Phrasal Verbs, Idioms and Proverbs, Homonyms and Homographs; Lexical Range, Word Games		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Communicate with people using English language functions including greetings, introductions, making and responding to requests, suggestions, invitations and apologies, conducting simple transactions in shops and offices, asking for and giving directions, etc
CO2	Go through a text and identify specific and global information.
CO3	Become more knowledgeable about speaking strategies and speak effectively using appropriate words, expression, tone and pronunciation.
CO4	Be aware about various reading strategies and read and comprehend academic and non-academic prose (text).
CO5	Write systematically using nuances of writing
CO6	Express their opinion and likes and dislikes, advice and convince others in a more polite and accepted way.

Evaluation Scheme:

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Recommended Study Material:

❖ Text book:

1. Sanjay Kumar and PushpLata (Second Edition, 2015), Communication Skills, Oxford University Press, New Delhi
2. M V Rodriques (2013), Effective Business Communication, Concept Publishing Company (P) Ltd., New Delhi
3. Krishna Mohan and Meera Banerji (2010), Developing Communication Skills, Macmillan Publications India Ltd., New Delhi

❖ Reference book:

1. Mohan and Meenakshi Raman (2006), Effective English Communication, McGraw-Hill Publishing Company Limited, New Delhi
2. Geoffrey Leech & Jan Swartvik (1994), A Communicative Grammar of English, Longman Publications, New York
3. Jones Leo (1979), Functions of English, Cambridge University Press, UK
4. European Journal of Language and Literature Studies Vol.1 Nr, 1 April 2015
5. English for Academic Purpose: A Tool for Enhancing Students' Proficiency in English Language Skills

❖ Web material:

1. <https://www.futurelearn.com/courses/language-assessment>
2. <https://www.coursera.org/learn/importance-of-listening?#syllabus>
3. <https://www.futurelearn.com/courses/english-academic-study>
4. <https://www.coursera.org/learn/grammar-punctuation>
5. <https://www.coursra.org/specializations/improve-english>

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Two-credit course

Course Code: CUUV101

Course Title: Community Engagement and Sustainable Development (CESD)

Credits	Course Duration (Hours)	Course Span
2	45	One semester (1 st / 3 rd)

Mode	Week	Hours	Sessions
Interactive Mode	15	30	15
Practical mode	1(2 Days)	10	-
Reflective Mode	5	5	5
Total	45	45	20

Session Planning for CESD in Three Modes

1. Interactive Mode

- **Duration:** 30 hours
- **Activities:**
 - Interactive dialogues
 - Group discussions
 - Role-playing scenarios
 - Expert lectures
- **Objectives:**
 - Enhance understanding of SDGs and community engagement.
 - Develop communication and management skills.

2. Practical Mode

- **Duration:** 10 hours
- **Activities:**
 - Fieldwork and community visits
 - Civic, social, nature, health, and education awareness programs
 - Research and surveys
- **Objectives:**
 - Apply theoretical knowledge to real-world scenarios.
 - Engage directly with communities and understand their needs.

3. Reflective Mode

- **Duration:** 05 hours
- **Activities:**
 - Case studies analysis
 - Report writing and submissions
 - Reflection sessions
- **Objectives:**
 - Analyze and document experiences and learnings.
 - Reflect on the impact of community engagement activities.

This structured approach ensures comprehensive learning, practical experience, and reflective thinking, equipping students with the necessary skills for community engagement and sustainable development.

1. Interactive Mode

Students will participate in a variety of activities such as personal hygiene sessions, cleanliness drives, involvement in the Fit India Movement, awareness campaigns, creative tasks like poster making and slogan writing, essay competitions, discussions under the theme Let's Talk Green, organizing awareness rallies and others.

Sr.No	Week	List of Activities	Duration
1	1 Week	Webinar on Personal Hygiene	2 Hours
2	2 Week	Seminar for fit India Movement	2 Hours
3	3 Week	Youth survey for National Youth Policy	2 Hours
4	4 Week	Expert Session on voters Awareness	2 Hours
5	5 Week	Seminar on water resources management	2 Hours
6	6 Week	Maths Science Model Training	2 Hours
7	7 Week	Poster Making , Slogan Writing, Essay Writing sessions	2 Hours
8	8 Week	Art & Craft Training to children of rural Area	2 Hours
9	9 Week	Webinar on 'Let's Talk Green'	2 Hours
10	10 Week	Webinar on Effective use of PPE Kit	2 Hours
11	11 Week	Cleanness Drive on Campus	2 Hours
12	12 Week	Expert lecture on Civic Awareness	2 Hours
13	13 Week	Awareness Rally on Addiction	2 Hours
14	14 Week	Street Play on Social Issues	2 Hours
15	15 Week	Tree Plantation	2 Hours

- Above mentioned are suggested topics, respective PO- CC may add upon as per their need and execute accordingly

2. Practical Mode

Sr.No	Six Days Workshop	Hours
1	Cleanness Drive in Villages, Education Amelioration Activities in Rural Schools, Awareness Drive on diseases	5 Hours
2	NO Plastic Use Awareness Drive –Rally etc. Education Awareness by door to door Campaign, Survey for government schemes	5 Hours

3. Reflective Mode

Sr.NO	Self-Development Skill	Hours
1	Assignments & Reports For All Above Activities , Self-guided Activities	05 Hours

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Two-credit course

Course code: - CUUV102
Physical Education and Sports

Credits	Course Duration (Hours)	Course Span
2	45	One semester

Mode	Week	Hours	sessions
Regular mode	15	30	15
Workshop mode	2 Days	10	-
Self-learning	1 day	05	-
Total	15 (3days)	45	15

➤ **TEACHING LEARNING STRATEGIES**

The content will be taught by using lecture, demonstration, explanation, presentation methods, videos, learning by doing, Whole part whole method, Drills

➤ **COURSE LEARNING OUTCOME**

After completing this course, the students will be able to

- Acquire, analyse and interpret basic skills
- Appraise the rules and regulation.
- Exhibit and assess a range of fundamental abilities, method and gaming strategies.
- Improve their physical fitness, including aspects such as strength, speed, agility, and endurance, to enhance their performance on the field.
- Enhance their ability to make effective decisions under pressure, including when to during the game.
- Understand the importance of sportsmanship, respect for opponents, referees, and teammates, and adherence to fair play principles.
- Learn mental preparation techniques to enhance focus, concentration, confidence, and resilience in competitive situations.
- Learn techniques for injury prevention, including proper warm-up, cool- down, and recovery strategies.

One Semester Course (Regular mode)

PLAN NO	SESSION TOPIC Football	SESSION TOPIC Basketball	SESSION TOPIC Volleyball	SESSION TOPIC Kabaddi
1	Football fundamental/Passing	Fundamental Basketball /Passing & Receiving	Volleyball fundamental/passing	footwork and agility training
2	Receiving/Controlling	Dribbling & Ball Handling	Service training	Simple raid
3	Dribbling	Controlling & Dribbling	Setting	Bonus point
4	Shooting	Passing & Catching	Blocking	Running hand-touch
5	First touch	Shooting / Jump Shot	Attack training	Toe touch
6	Passing and Supporting	Lay ups Shot	Defensive/offensive skill training	Kick training
7	Long ball /Air ball	Rebounding & Stealing	Footwork and agility training	Cover block
8	Linkup and Coordination	Defending (Defense)	Advance digging training	Single block
9	Set piece (penalty kick, corner kick) (free kick) (Throw in)	Attacking (Offense)	Position-specific drill training	Ankle hold
10	Dribbling (1vs1) (1vs2)	Off-ball Movements	“w” formation / 5-1 Defence system	Back hold
11	One touch passing	Two-on-Two and Three-on-Two	Team drills –rotational drills	Knee hold
12	Defending	Blocking & Intercepting	Combination drills session	Double thigh-hold
13	Attacking	Inbounding the Ball	Tactical session	Team drills-communications, rotational
14	Weight of the ground pass	Fast Break	Peak performance	Kabaddi communication drills
15	Shadow game(match style)	Tactical & Technical Skills	Tournament preparation	Game like drills

Workshop mode- Two Days

Workshop Duration	Football	Basketball	Volleyball	Kabaddi
Two Days	Official match	Official match	Official match	Official match
	Official match	Official match	Official match	Official match

Self-learning mode

Course Period	SELF -LEARNING MODE	Particulars
1 semester	Football	Case Studies or Project of Matches/ Players / World Tournaments /Book/Survey
	Basketball	
	Volleyball	
	Kabaddi	

A self-learning this course aims to develop a range of knowledge areas to improve students' understanding and performance in the sport. Here are the potential outcomes:

- Self-Paced Learning
- Resource Utilization
- Better Psychological Skills
- Enhanced Teamwork and Communication
- Knowledge of Rules and Regulations

This course execution Sports Faculty: -

Games	Football	Basketball	Volleyball	Kabaddi	Yoga
Faculty	Harsh Parmar	Ajay Raval	Pinkal Gamit	Tejal Gamit	External expert
Venue	Football Ground	Basketball court	Volleyball Ground ,New Sports Complex	Kabaddi Ground , New Sports Complex	

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Two-credit course

Course Code: - CUUV103
National Cadet Corps (NCC)

Credits	Course Duration (Hours)	Course Span
2	90	During Initial Two years of study at Under Graduate Level

Course Description

This course aims to develop leadership, discipline, and patriotism among students through participation in the National Cadet Corps. It covers theoretical and practical aspects of NCC training. The course intends to produce responsible, disciplined citizens with spirit of righteousness to contribute in the development of the nation.

Prospective Students

Undergraduate students of CHARUSAT interested in National Cadet Corps (NCC).

Course Objectives

1. To develop leadership skills among students.
2. To instill a sense of discipline and teamwork.
3. To promote physical fitness and a spirit of adventure.
4. To create awareness about national security and social issues.
5. To develop patriotism, feeling of love, devotion, and a sense of attachment to the nation.

Course Outline

Components	Description
Introduction to NCC	History and objectives of NCC. NCC organization and structure.
Leadership and Discipline	Principles of leadership. Importance of discipline in NCC and life.
Drill and Weapon Training	Basic drill movements. Introduction to weapons used in NCC.
Social Service and Community Development	Role of NCC in community development. Participation in social service activities including management of disaster, election etc.
Adventure Activities	Trekking, camping, and rock climbing. Importance of adventure in personal development.
National Security and Defense	Basics of national security and defense.

Course Delivery

The course content shall be delivered through a combination of theoretical, practical and experiential approach fostering both philosophical and practical skills. The course shall be transacted through discussions, practical, drills, lectures from experts in the field. Students shall be required to actively participate in the activities and complete the assigned tasks.

The course shall be implemented by a team of ANO/CTO –NCC Unit CHARUSAT. This structure ensures effective coordination and supervision of the course activities, leveraging the expertise and resources for the maximum impact.

Course Assessment

The assessment shall be carried out through active and consistent participation in NCC activities, practical exams and participation as a team member in various NCC activities. Student who qualify for NCC B certificate exam will be exempted from certain assessment requirements.

Course Outcome

On completion of the course, students will be able to:

- Develop leadership skills among students and society.
- Exhibit social service characteristic.
- Understand the importance of national security and defense.

The course intends to provide a platform to the students to exhibit their skills, knowledge, competence and quality to meet the expectations of the Nation from youth. This course will be implemented as per the guidelines of CHARUSAT as well as NCC Directorate.

FACULTY OF TECHNOLOGY AND ENGINEERING
M. S. PATEL DEPARTMENT CIVIL ENGINEERING

CLOV101 Environmental Sciences

A. CREDIT AND SCHEME:

Teaching Scheme	Theory/ Practical	Total	Credit	Evaluation Scheme				
				Theory		Practical		Total
Hours/week	2	2	2	Internal	External	Internal	External	
Marks	50	50		--	--	25	25	50

B. OUTLINE OF THE COURSE:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Multidisciplinary nature of environmental Sciences	03
2	Environmental Pollution	07
3	Ecology & Ecosystems	05
4	Biodiversity and Conservation	04
5	Natural Resources	04
6	Human Communities and the Environment	07

Total Hours (Theory): 00

Total Hours (Lab): 30

Total Hours: 30

C. DETAILED SYLLABUS:

1.	Multidisciplinary nature of environmental studies	03 Hours	11%
1.1	Definition, scope and importance		
1.2	Earths-Evolution and Systems		
1.3	Components of the Environment: Biotic, Abiotic, Atmosphere, Lithosphere, Hydrosphere, Biosphere		
1.4	Concept of sustainable development		
2.	Environmental Pollution	07 Hours	23%
2.1	Environmental pollution: types, causes, effects and controls; Air, water, soil and noise pollution		
2.2	Nuclear hazards and human health risks		
2.3	Solid waste management: Control measures of urban and industrial waste		
2.4	Pollution case studies – Ganga/Yamuna River, Bhopal Gas Tragedy, Delhi Air Pollution, Effect of Pandemics on the Environment		
3.	Ecology & Ecosystems	05 Hours	17%
3.1	Concept of an ecosystem, Structure and function of an ecosystem		
3.2	Producers, consumers and decomposers		
3.3	Energy flow in the ecosystem, Food chains, food webs and ecological pyramids		
3.4	Case studies of the following ecosystems: Forest ecosystem, Grassland ecosystem Aquatic ecosystems (ponds, streams, lakes, rivers)		
4.	Biodiversity and Conservation	04 Hours	13%
4.1	Introduction – Definition: genetic, species and ecosystem diversity		
4.2	Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values		
4.3	Hot-spots of biodiversity, Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts.		
4.4	Endangered and endemic species of India, Conservation of Biodiversity Act 2002/ BD Rule 2004: Mandate & Functions of National Biodiversity Authority (NBA), Role of State Biodiversity Board (SBB) and Biodiversity Management Committees (BMC).		
4.5			
5.	Natural Resources	04 Hours	13%
5.1	Renewable and non-renewable resources		
5.2	Recyclable and Non-recyclable resources		
5.3	Energy resources: Growing energy needs, use of alternate energy sources. Case studies.		
5.4	Role of an individual in conservation of natural resources		
6.	Human Communities and the Environment	07 Hours	23%
6.1	Pandemics: Causes, Effects, Impact on the Environment: Positive & Negative, Lessons to learn		
6.2	Floods, Cyclones, Earthquakes, Landslides & Forest Fires		
6.3	Human population growth: Impacts on environment, human health and welfare.		
6.4	Case Studies: Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents, etc.		

A. INSTRUCTIONAL METHOD AND PEDAGOGY:

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

EVALUATION:

There will be end semester university examinations based on Practical/ Viva. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 25 marks for internal evaluation and 25 marks for external evaluation. The concerned teacher shall evaluate students and distribute the marks

Evaluation Scheme

The students' performance in the course will be evaluated through the following components:

Component		Marks
Internal	Attendance	05
	Assignment/ Individual Activity Participation/ Quiz, etc.	10
	Quiz	10
External	Group Activity Participation, Project Report, Case Study with Report, Online Certification Course, MOOCS, Field Work with report, Presentation etc.	10
	Viva/ Oral Examination	15
Total		50

Note: In internal component assignment and quiz are compulsory but in external component of 10 marks each subject teacher is free to conduct any activity mentioned above (that particular activity should be same for whole class).

Course Outcomes (COs):

At the end of the course, students will be able:

CO1	To perceive the elementary knowledge about natural environment and its relation with science.
CO2	To identify and analyze human impacts on the environment.
CO3	To understand the facts and concepts of natural and energy resources thereby applying them to lessen the environmental degradation.
CO4	To Initiate new and innovative environmental friendly practices.

CO5	To communicate on recent environmental problems thereby creating awareness among society.
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Recommended Study Material:

Text Books:

1. Varandani, N.S., Basics of Environmental Studies
2. Sharma, J. P., Basics of Environmental Studies

Reference Books:

1. Shah Shefali & Goyal Rupali, Basics of Environmental Studies
2. Agrawal, K.C., Environmental Pollution: Causes, Effects & Control
3. Dameja, S. K., Environmental Engineering & Management
4. Rajagopalan, R., Environmental Studies, Oxford University Press
5. Wright Richard T. & Nebel Bernard J., Environmental Science
6. Shah, S.G., Shah, S.G. & Shah, G. N., Basics of Environmental Studies, Superior Publications, Vadodara

Web Materials:

- <http://nptel.iitm.ac.in/courses/Webcourse-contents/IITDelhi/Environmental%20Air%20Pollution/index.htm>
- <http://nptel.iitm.ac.in/video.php?subjectId=105104099>
- [http://apollo.lsc.vsc.edu/classes/met130/notes/chapter1/vert temp_all.html](http://apollo.lsc.vsc.edu/classes/met130/notes/chapter1/vert_temp_all.html)
- <http://www.epa.gov>
- <http://www.globalwarming.org.in>
- <http://nopr.niscair.res.in>
- <http://www.indiaenvironmentportal.org.in>
- <http://nbaindia.org/>

Bachelor of Business Administration (BBA) – Business Analytics Programme

SYLLABI

(Semester – II)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

BBU151: FUNDAMENTALS OF ECONOMICS
YEAR 1, SEMESTER – II

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Knowledge of Economics enables students to monitor and study changes occurring in specific countries or individual sectors of an economy. It deals with the optimum allocation of scarce resources in a manner that minimizes the cost. Economics enables the students to have an insight into how it works at the micro as well as macro levels. Understanding economics helps them to make decisions about using scarce resources and how global, as well as domestic economic changes, will impact businesses. It enables us to understand people, businesses, markets, and governments, and therefore better respond to the threats and opportunities that emerge when things change.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Economics	10
2	Demand and Supply Analysis	10
3	Production, Cost and Revenue Analysis	12
4	Pricing in Different Market Structures	10
5	Introduction to Macroeconomics	10
6	Contemporary Issues / Affairs: Money, Banking and Macroeconomic Indicators	08
Total Contact Hours		60

Course Syllabus

Module 1: Introduction to Economics

10 hours

Definition, Nature and Scope of Economics, Micro and Macro Economics, Role of Economics in Decision Making, Nature and Scope of Economics, Deductive and Inductive, Method of Economics, Positive and Normative Economics, Basic Problems of an Economy, Production Possibility Frontier, Budget Line, Types of Economic System

Module 2: Demand and Supply Analysis

10 hours

Meaning of Demand, Its Types, Law of demand & its determinants, Demand Function, Elasticity of demand-price, Elasticity of demand, Income, Elasticity of demand, Cross, Elasticity of demand, Law of Supply, Supply Schedule & Supply Curve, Price elasticity of supply

Module 3: Production, Cost and Revenue Analysis

12 hours

Production function, Types of Production Function, Law of Returns – Law of Variable Proportions, Law of Increasing Returns, Law of Constant Returns, Law of Diminishing returns, Returns to scale, Elements of Cost, Relationship between Production and Cost Average and Marginal cost curves, Relationship between average and marginal cost, Concept of revenue, Revenue Curve, Relationship between average and marginal revenue

Module 4: Pricing in Different Market Structures

10 hours

Different market structures – Perfect Competition, Imperfect Competition, Monopolistic Market, Oligopoly Market, and Duopoly Market, Price determination in Perfect Competition, Monopoly, Monopolistic and Oligopoly Markets

Module 5: Introduction to Macroeconomics

10 hours

Meaning, Definition and Importance of Macro Economics, National Income: Concept and Its Measurement, Limitations in Measurement of National Income, Real GDP and Nominal GDP

Module 6: Contemporary Issues / Affairs: Money, Banking and Macroeconomic Indicators

08 hours

Interest rate, Credit growth and Inflation (CPI and WPI), RBI and its Role, Monetary Policy and Fiscal Policy, IIP

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 2 hours based on written paper for 50 marks.

Text Book:

1. Mankiw, Principles of Microeconomics, Cengage Learning, Latest Edition.

Reference Books:

1. Dwivedi, D.N, Managerial Economics, Vikas Publishing House Pvt. Ltd, Latest Edition.
2. M.L. Jhingan & B.K. Jhingan, Principles of Economics, Vrinda Publications Ltd.
3. Varshney and Maheshwary, Managerial Economics, Sultan Chand & Sons, Latest Edition.
4. Gupta, G.S, Managerial Economics, McGraw Hill, Latest Edition
5. Mehta, P.L., Managerial Economics: Analysis, Problems and Cases, Sultan Chand & Sons, Latest Edition.
6. David Dranove, David Besanko, Mark Shanley, Scott Schaefer, Economics of Strategy, Wiley publications
7. R. Glenn Hubbard, Anthony Patrick O'Brien, Macroeconomics, Pearson Education

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Build the ability to understand the basics of Economics.
CO2	Comprehend the function of demand and supply and how elasticity is measured with demand and supply.
CO3	Developing the ability in understanding various market structures and their price determination.
CO4	Equip the students to gain knowledge about macroeconomic indicators that would affect the functioning of the businesses
CO5	Gaining knowledge about production, cost and revenue structure involved in producing goods and services.
CO6	Understanding the concept of measuring the value of economic activities

Course Outcomes Mapping

Module	Module Name	Course Outcomes					
		CO1	CO2	CO3	CO4	CO5	CO6
1	Introduction to Economics	√					
2	Demand and Supply Analysis		√				
3	Production, Cost and Revenue Analysis					√	
4	Pricing in Different Market Structures			√			
5	Introduction to Macroeconomics						√
6	Contemporary Issues / Affairs: Money, Banking and Macroeconomic Indicators				√		

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross-Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	Build the ability to understand the basics of Economics	2	3	2	1
CO2	Comprehend the function of demand and supply and how elasticity is measured with demand and supply	2	2	2	1
CO3	Developing the ability in understanding various market structures and their price determination	1	2	2	1
CO4	Equip the students to gain knowledge about macroeconomic indicators that would affect the functioning of the businesses	1	3	2	1
CO5	Gaining knowledge about production, cost and revenue structure involved in producing goods and services	2	3	2	1
CO6	Understanding the concept of measuring the value of economic activities	1	3	2	1

**BBU152: INTRODUCTION TO DATA SCIENCE
YEAR 1, SEMESTER – II**

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

The data is the most important element of any analysis and its interpretation for effective decision making in an organization. This course covers topics needed to solve problems involving data, which includes data science process which has the steps which are preparation collection and integration, characterization and presentation through visualization, analysis using machine learning and data mining and applications.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Data Science	10
2	The Data Science Process	10
3	Introduction to Machine Learning	10
4	Text Mining and Text Analytics	10
5	Data Visualization	10
6	Contemporary Issues	10
Total Contact Hours		60

Course Syllabus

Unit – 1: Introduction to Data Science

10 hours

Benefits and uses of data science, Facets of data, The data science process, The big data ecosystem

Module 2: The Data Science Process

10 hours

Defining research goals and creating a project charter, Retrieving data, Cleansing, integrating, and transforming data, Exploratory data analysis, Build the models, Presenting findings and building applications

Module 3: Introduction to Machine Learning

10 hours

Applications, The modelling process, Types of machine learning, ML tools

Module 4: Text Mining and Text Analytics

10 hours

Text mining in the real world, Text mining techniques, Bag of word, Stemming and lemmatization, Decision tree classifier, Text mining tools

Module 5: Data Visualization

10 hours

Data visualization options, Cross filter, Interactive dashboards, Dashboard development tools

Module 6: Contemporary Issues

10 hours

Case Studies, Model Development

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Text Book:

1. Introduction to Data Science, Davy Cielen, Arno D. B. Meysman, Mohamed Ali, Dreamtech Press (Latest Edition)

Reference Books:

1. Introduction to Data Science, Rafael A. Irizarry, CRC Press.

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Develop a clear understanding of the data science process for effective decision making.
CO2	Understand the tools and techniques available to learn the data science process for the model development.
CO3	Identify data analysis through modeling using machine learning and data mining techniques.
CO4	Create and modify tools for data analysis and visualization of the data for decision making.

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction to Data Science	√			
2	The Data Science Process		√		
3	Introduction to Machine Learning			√	
4	Text Mining and Text Analytics			√	
5	Data Visualization				√
6	Contemporary Issues				

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	Develop a clear understanding of the data science process for effective decision making.	3	2	2	1
CO2	Understand the tools and techniques available to learn the data science process for the model development.	1	2	2	3
CO3	Identify data analysis through modeling using machine learning and data mining techniques.	2	2	2	3
CO4	Create and modify tools for data analysis and visualization of the data for decision making.	1	2	3	3

BBU153: BASICS OF QUANTITATIVE METHODS
YEAR 1, SEMESTER – II

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Theory	04	60

Rationale

Quantitative techniques or methods are of paramount importance in the business world and Management. Where our success entirely depends on our ability to take correct and timely decisions. Today, most of the business problems have found their quantitative representation, which makes this field of study more interesting and useful to the students and businesspersons alike. These methods are a great aid to the business strategists. In the present world, knowledge of these methods is a prerequisite to enter any business environment.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Understanding of Fundamentals	08
2	Probability and Probability Distribution	12
3	Measure of Central Tendency	12
4	Testing of Hypothesis: Large Sample Tests	12
5	Correlation and Regression Analysis	12
6	Contemporary Issues / Affairs	04
Total Contact Hours		60

Course Syllabus

Module 1: Understanding of Fundamentals

08 hours

Definition of Statistics with Data, Variables and its types, Functions and Limitations, Scientific Approach to Quantitative techniques, Probability Distributions, Classification and Tabulation of Statistical data, Types of diagrams, Graphs and Charts

Module 2: Probability and Probability Distribution

12 hours

Introduction of Probability, Structure of Probability, Marginal, Union, Joint and Conditional Probability, Discrete Distribution: Binomial and Poisson Distribution, Continuous Distribution: Normal Distribution

Module 3: Measure of Central Tendency

12 hours

Meaning, Understanding the Mean, Median and Mode, Measures of Dispersion: Variance, Geometric Mean and Harmonic Mean, Application of measures of central tendency in Management

Module 4: Testing of Hypothesis: Large Sample Tests

12 hours

Different definitions and its Meaning, Testing of Hypothesis for LST, Small Sample Test: t test, F test

Module 5: Correlation and Regression Analysis

12 hours

Introduction of Correlation, Types of Correlation, Methods to find Correlation, Scatter Diagram, Karl Pearson's Correlation Coefficient, Spearman Correlation Coefficients, Introduction of Regression, Regression Coefficients, Properties of Regression Coefficients, Regression Equations Meaning: Multiple Regression Equation

Module 6: Contemporary Issues / Affairs

04 hours

Meaning, Definition, Functions and concepts of Data Analytics, Tools of Data Analytics, Issues related with Data Analytics

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 2 hours based on written paper for 50 marks.

Text Book:

1. Black, K., Business Statistics for Contemporary Decision Making (Latest Edition), Wiley Publishers

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understanding the Basic knowledge of Statistical Terms
CO2	Analyzing the Types of the Data and Variables with Applications
CO3	Competence in Decision Making
CO4	Applying Probability and Distributions
CO5	Developing Inferential Statistics and its Application in field work.

Course Outcomes Mapping

Module	Module Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Understanding of Fundamentals	√				
2	Probability and Probability Distribution	√	√			
3	Measure of Central Tendency		√			
4	Testing of Hypothesis: Large Sample Test	√	√	√		
5	Correlation and Regression Analysis	√	√		√	
6	Contemporary Issues / Affairs					√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understanding</i> the Basic knowledge of Statistical Terms	2	1	3	3
CO2	<i>Analyzing</i> the Types of the Data and Variables with Applications	2	1	3	3
CO3	<i>Competence</i> in Decision Making	3	1	3	3
CO4	<i>Applying</i> Probability and Distributions	2	1	2	3
CO5	<i>Developing</i> Inferential Statistics and its Application in field work.	2	1	1	1

BBU154: INTRODUCTION TO COMPUTING
YEAR 1, SEMESTER – II

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
04	Practical	04	60

Rationale

This course will enable the students gain sufficient knowledge and experience to programming concepts. The course will help the students understand python environment, its modules and libraries.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Python Programming	10
2	Control Statements	10
3	Functions	10
4	Manipulating Strings	10
5	Files and Exceptions	10
6	Contemporary Issues	10
Total Contact Hours		60

Course Syllabus

Unit – 1: Introduction to Python Programming

10 hours

Introduction, Variables and assignment statements, Arithmetic, Print Function, Decision making, Introduction to data science: Descriptive

Module 2: Control Statements

10 hours

Introduction, if statement, if...else and if..elif...else statements, while statement, for statement, Augmented assignment, break and continue statements, Boolean operators and, or and not, Measures of central tendency, Intro to data science: Measures of central tendency

Module 3: Functions

10 hours

Introduction, Defining functions, Functions with multiple parameters, Random-number generation, Math module function, Intro to data science: Measures of dispersion

Module 4: Manipulating Strings

10 hours

Introduction, Formatting strings, Concatenating and repeating strings, Stripping whitespace from string, Searching substring, Replacing strings, Splitting and joining strings, Intro to data science: Pandas and data munging

Module 5: Files and Exceptions

10 hours

Introduction, Files, Text-file processing, Updating text files, Handling exceptions, finally clause, Intro to data science: working with CSV files

Module 6: Contemporary Issues

10 hours

Contemporary issues in analytics that can be analyzed and solved using Python

Course Evaluation

I. Continuous and Comprehensive Evaluation (50 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (50 Marks):

- University Examination will be carried out at the end of the semester.
- Practical examination will be based on project or report presentation and viva-voce examination or software based analysis depending on the nature of the course.

Text Book:

1. Python for Programmers, Paul Deitel and Harvey Deitel, (Latest Edition), Pearson.

Reference Books:

1. Data Analytics using Python, Bharti Motwani, (Latest Edition), Wiley.

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the foundation of programming
CO2	Learn the control structure in programming
CO3	Use the modules and libraries to develop program for problem solving
CO4	Understand the file handling operations in python

Course Outcomes Mapping

Module	Module Name	Course Outcomes			
		CO1	CO2	CO3	CO4
1	Introduction to Python Programming	√			
2	Control Statements		√		
3	Functions			√	
4	Manipulating Strings			√	
5	Files and Exceptions				√
6	Contemporary Issues				

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understand</i> the foundation of programming	1	1	1	1
CO2	<i>Learn</i> the control structure in programming	2	2	3	3
CO3	<i>Use</i> the modules and libraries to develop program for problem solving	2	3	3	3
CO4	<i>Understand</i> the file handling operations in python	2	3	3	3

BBU155: BUSINESS CASE ANALYSIS
YEAR 1, SEMESTER – I

Credit	Component	Instruction/ Contact/ Practical Hours (Per Week)	Instruction/ Contact/ Practical Hours (Per Course)
02	Theory	02	30

Rationale

Bridging theoretical knowledge with practical application is an integral part of management studies. By analyzing real-world case studies, students develop decision-making and problem-solving abilities essential for future business leaders. This approach cultivates critical thinking by challenging assumptions and evaluating evidence, while also enhancing communication skills through group discussions and presentations. Exposure to diverse business scenarios broadens students' understanding of the complexities within different industries, preparing them to tackle dynamic challenges they will encounter in their careers. The subject empowers the students to analyze, strategize and communicate effectively in the business world.

Course Pedagogy

The course will emphasize self-learning and active classroom interaction based on students' prior preparation. The course instructor is expected to prepare a detailed session-wise schedule, showing the topics to be covered, the reading material and case material for every session. Wherever the material for any session is drawn from sources beyond the prescribed text-book, reference books, journals and magazines in the library, or from websites and other resources not accessible to the students, the course instructor should make the material available to the students well in advance, so that the students can come prepared for the classes. The pedagogical mix will consist of classroom sessions, assignments, presentations, management exercises or games/simulations or activities and student feedback.

Course Outline

Module Number	Title of the Module	Contact Hours
1	Introduction to Case Analysis	05
2	Process and Preparation of Case Analysis	05
3	Case Analysis Tools I	05
4	Case Analysis Tools II	05
5	Developing Possible Solution for Case	05
6	Writing A Business Case Study	05
Total Contact Hours		30

Course Syllabus

Unit – I Introduction to Case Analysis **05 hours**

Types of Cases, Working with Cases, researching for the case, Approaches/Methods for Case Analysis, Learning from the Case.

Module 2: Process and Preparation of Case Analysis **05 hours**

Identifying the business/managerial Problem, Process of Performing Case Analysis, Written/Oral Case Analysis, Exhibits of Case Analysis, Commandments of Case Analysis.

Module 3: Case Analysis Tools I **05 hours**

SWOT Analysis, TOWS Analysis, PESTLE Analysis.

Module 4: Case Analysis Tools II **05 hours**

Porter's Five force model, Value-Chain Analysis Model, Cause-Effect Diagrammatic Model, 7 QC Tools (Check Sheet, Fishbone Diagram, Histogram, Pareto Chart, Control Chart, Scatter Diagram, Stratification Diagram).

Module 5: Developing Possible Solution for Case **05 hours**

Evaluation of Alternatives, Selection of Appropriate Solution, Priority based solution.

Module 6: Writing A Business Case Study **05 hours**

Steps involved in Writing a Business Case Analysis

Course Evaluation

I. Continuous and Comprehensive Evaluation (25 Marks)

Component	Marks
Quiz	70
Assignment/ Project/ Presentation	50
Online Course	60
Internal Examination	40
Participation	30
Total marks (CCE)	250

II. Semester End Evaluation (25 Marks):

- University Examination will be carried out at the end of the semester.
- Theory examination will be for 1 hour based on written paper for 25 marks.

Text Book:

1. James W. Brannock, BCA: Business Case Analysis: Examples, Concepts & Techniques: Featuring the 8-day BCA, STS Publication (Latest Edition)

Reference Books:

1. Dr. Ramesh R Kulkarni, Dr. Shrinivas R Patil, Rajashekhar R Navalagi, Rangappa K Yaraddi, Management Case Studies: A Student's Handbook, Notion Press, (Latest Edition)
2. Peter Drucker, Management Cases, Harper Collins, (Latest Edition)
3. William Ellet, The Case Study Handbook, how to Read, Discuss, and Write Persuasively about Cases, Harvard Business Press, (Latest Edition)
4. Charles Hill, Gareth Jones, Strategic Management Theory: An Integrated Approach, CENGAGE Learning (9th Edition)
5. J. David Hunger, Thomas Wheelen, Essentials of Strategic Management, Pearson Publications (5th Edition)

Programme Outcomes: Upon successful completion of the course, students will be,

PO1	Analyze contemporary issues and apply management knowledge for resolution.
PO2	Practice effective communication and interpersonal skills.
PO3	Understand Global Perspectives and exhibit Cross-Cultural sensitivity.
PO4	Demonstrate Attitude for continuous learning and development.

Course Outcomes: Upon successful completion of the course, students will be,

CO1	Understand the key aspects of a business case and its analysis.
CO2	Identify business and managerial problems.
CO3	Demonstrate competence in applying strategic analysis tools.
CO4	Evaluate alternative solutions and select the most appropriate ones.
CO5	Apply the understanding of business cases in writing a business case study.

Course Outcomes Mapping

Module	Module Name	Course Outcomes				
		CO1	CO2	CO3	CO4	CO5
1	Introduction to Case Analysis	√				
2	Process and Preparation of Case Analysis		√			
3	Case Analysis Tools I			√		
4	Case Analysis Tools II			√		
5	Developing Possible Solution for Case				√	
6	Writing A Business Case Study					√

Course Articulation Matrix

		PO1	PO2	PO3	PO4
		<i>Analyze</i> contemporary issues and apply management knowledge for resolution.	<i>Practice</i> effective communication and interpersonal skills.	<i>Understand</i> Global Perspectives and exhibit Cross- Cultural sensitivity.	<i>Demonstrate</i> Attitude for continuous learning and development.
CO1	<i>Understand</i> the key aspects of a business case and its analysis.	1	2	1	1
CO2	<i>Identify</i> business and managerial problems.	3	2	1	3
CO3	<i>Demonstrate</i> competence in applying strategic analysis tools.	1	2	2	1
CO4	<i>Evaluate</i> alternative solutions and select the most appropriate ones.	1	2	1	1
CO5	<i>Apply</i> the understanding of business cases in writing a business case study.	1	1	1	3

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester-II

HSUS101: PAINTING

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
				Contact Hours/Week	Internal	External	Internal	External	
II	HSUS101	Painting	02	02	--	--	25	25	50

II. Course Objectives

- To encourage/ foster creativity among the students
- To introduce students to the fundamental processes of visual perception and artistic expression
- To cultivate / spawn awareness among students about the significance of arthistory, art criticism and aesthetics
- To help the students understand the meanings of concept, designs, shapes, colors, medium, and format
- To give the students the firsthand experience of design, painting, colors, light, shapes, shades and other important aspects of painting
- To develop in students an understanding of major styles and contemporary issues in painting

III. Course Outline

Module No.	Title	Classroom Contact Sessions
1	An Introduction to Painting • <i>An Introduction to Painting</i> • <i>Principles of Composition</i> • <i>Medium and Techniques of Painting</i> • <i>History of Painting: Folk Indian Painting /Western Painting</i> • <i>2D and 3D Painting</i>	2
2	Drawing from Nature and Object • <i>Objects of Drawing: Nature and Manmade /Artificial Objects</i> • <i>Drawing Still / Live Objects</i> • <i>Drawing from Memory</i> • <i>Drawing from Life</i>	4
3	Colour Design and Colour Value • Color Theory: • <i>Color wheel (primary/secondary,</i>	6

Module No.	Title	Classroom Contact Sessions
	<i>complementary), transparency/opacity, hue, value (intensity, brightness), chroma (saturation, purity) & temperature (warm/cold)</i> Color Contrast & Attributes: • <i>Interaction, harmony, psychology/mood, culture & expression</i> Media Characteristics & Surfaces: • <i>Acrylic, oil, paper, wood & canvas (primed/unprimed)</i>	
4	Composition and Perspective Composition: • <i>Space, movement, balance, asymmetry, rhythm, shapes, proportion & lighting</i> • <i>Perspective: An approximate reproduction</i> Types of Perspectives: • <i>Linear Perspective, One-point Perspective,</i> • <i>Two-point Perspective</i> • <i>Three-point Perspective,</i> • <i>Four-Point Perspective</i>	6
5	Figure Drawing and Proportion • <i>Proportions of the Human Body</i> • <i>Three views – Anterior (front), Lateral (side) and Posterior (back)</i> • <i>Fundamental Proportion – The Big Three</i>	4
6	Sketching • <i>Sketching and Freehand</i> • <i>Sketching Techniques</i> • <i>Sketch and Drawing Medium</i>	4
7	Contemporary Issues in Painting • <i>Contemporary Indian Art</i> • <i>Pioneers of Contemporary Indian Art</i> • <i>Contemporary Issues in Painting</i>	4
Total Hours		30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

V. Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- ✓ Formative: Internal / Continuous and Comprehensive Evaluation(CCE) – 25 Marks
- ✓ Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

VI. Major Learning Outcomes

At the end of the course, a student will

- i. have cultivated a sense of creativity.
- ii. be appreciative of art history, art criticism and aesthetics.
- iii. be able to recognize the elements of arts in painting.
- iv. have better cognizance and association of meaning of colors, shapes, and composition.
- v. be able to acknowledge the principles of painting as in design and colors, concept, medium and formats.
- vi. have instantaneous painting experience about designing, lights, shades and colors and such other important aspects.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester-II

HSUS102: PHOTOGRAPHY

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				
					Theory		Practical		Total
					Internal	External	Internal	External	
II	HSUS102	Photography	02	02	--	--	25	25	50

II. Course Objectives

- To introduce students to the tools and techniques of photography
- To provide students a thorough understanding of the mechanism and operations of a camera and help them understand the importance of aperture, shutter speed, film speed, depth of field, movement, and light meters to create a master shot
- To enable students to come out with a final project that demonstrates a single or a unified photographic idea or technique
- To explain students varied types of photographic representation including appropriation, persona, mixed media, non-objective images and engage them into experimentation using digital media
- To make students create a portfolio demonstrating creative uses of artificial and mixed lighting situations

III. Course Outline

Module No.	Title	Classroom Contact Sessions
1	An Introduction to Photography • Art, Design and Visualization • Basics of Photography and Various Types of Photography • Basics of Post Production • A Brief History of Photography: Early Experiments and Later Developments	03
2	Camera and Operating System • Role of Camera in the Photography • Types of Camera <i>Pin-hole, box, folding, large and medium format cameras, single lens reflex (SLR) and twin lens reflex (TLR), miniature, subminiature and instant camera</i> • Principal Parts of Photographic Camera <i>Lens, Aperture, Shutters, various types and their</i>	05

Module No.	Title	Classroom Contact Sessions
	<i>functions, focal plane shutter and in-between the lens shutter, shutter synchronization, self-timer</i> • <i>Types of Lenses</i> <i>Single (meniscus), achromatic, symmetrical and unsymmetrical lenses, telephoto, zoom, macro, supplementary and fish-eye lenses</i> • <i>Different Models of Camera, their Features and Operating Systems</i> • <i>Camera and Size of the Image, Speed and Power of Lens</i>	
3	Light and Shade • <i>Reflection and refraction of light</i> • <i>Dispersion of light through a glass prism, lenses</i> • <i>Colour Filters:</i> <i>Different kinds, Red, yellow, green, neutral density, half filters, filter factor, colour correction filter</i> • <i>Photographic Light Sources:</i> <i>Natural source, the Sun, nature and intensity of the sunlight at different times of the day, different weather conditions</i> • <i>Artificial light sources:</i> <i>Nature, intensity of different types of light sources used in photography namely; (i) Photo flood lamp, (ii) Spot light, (iii) Halogen lamp, Barn doors and snoot, lighting stands</i> • <i>Flash unit: Bulb flash and Electronic flash, main components, electronic flash units, studio flash, slave unit, multiple flash, computer flash, x-contact, exposure table</i>	10
4	Composition • <i>Different kinds of image formations</i> • <i>Principal focus and focal length of the lens</i> • <i>Depth of field, angle of view and perspective</i> • <i>Perspective and composition</i> • <i>Rules of composition</i>	09
5	Contemporary Issues in Photography • <i>Present Day Photography</i> • <i>Contemporary Photographers and their Contributions</i> • <i>Major Issues in Contemporary Photography</i>	03
	Total Hours	30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

V. Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- ✓ Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- ✓ Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

VII Learning Outcomes

At the end of the course, students will be able to

- Understand, appreciate and demonstrate innovative approach, beauty and acutecumen in the area of photography
- Develop photography skills and become familiar with the functions and importanceof the visual elements of nature and artificial objects
- Become independent thinkers who will contribute inventively and critically toculture through the making of art photography
- Have thorough understanding and acute sense of light and shade, composition, andpresentation of a piece of an art
- Experiment and Represent the cultivated sense and skills in Photography to themass
- Prepare an impressive portfolio encompassing holistic approach to art and other theareas of study.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
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Semester-II

HSUS103: SCULPTING

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
				Contact Hours/Week	Internal	External	Internal	External	
II	HSUS103	Sculpting	02	02	--	--	25	25	50

II. Course Objectives

- To promote creativity and aesthetic sense pertaining to Sculpting by introducing them to the history of sculpting, its basic concepts and contemporary techniques and issues
- To help the students understand and develop the skill of sketching and drawing from life, natural and manmade objects and structures using various means like pencil, pen, ink, crayon, chalk, Colour etc.
- To help them understand methods and materials of sculpture i.e. clay, plaster, cement, wood, stone, bronze, enlarging and reducing devices, welding torch etc.
- To help the students develop the sense of structure, and understand how forms achieve their structural unity through adherence to principles of physical nature of material being observed and studied (e.g. Plants, insects, minerals etc.)
- To introduce the basic visual elements of 2-D and 3-D designs with emphasis on fundamentals of two and three-dimensional designs
- To acquaint the learners with various perspectives to draw and mold a sculpture
- To make the learners understand the Colour theory and its practical usages
- To provide the students a sound background of the traditional and representational form in sculpture and enable them to develop their own vision

III. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Sculpture • <i>What Sculpture is and how it is different from other Arts</i> • <i>Basic elements, techniques, and history of sculpture</i> • <i>Form, Space and Basic Shapes in Sculpture</i> • <i>Casting materials, mold making, basic wood cutting and shaping, metal work, Welding, assembling, adhesives, plaster, mixing</i>	06
2	Fundamental Principles of Sculpture • <i>Basic Principles of Aesthetics in Sculpture</i> • <i>Visual Principles – balance, sequence, weight, and structural dynamics in sculpture</i> • <i>Structural Principles and communicative possibilities of sculpture</i>	10
3	Process of Modeling • <i>Additive and reductive processes</i> • <i>Major Techniques of Sculpture: Modeling, Carving, Pointing</i> • <i>Materials used in Modeling</i> • <i>Clay Modeling and Carving</i>	11
4	Contemporary Issues in Sculpture • <i>Sculpture and Present Day Context</i> • <i>Contemporary Sculptors and their Contribution</i> • <i>Major Contemporary Issues in Sculpture</i>	03
Total Hours		30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

V. Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- ✓ Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- ✓ Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

* Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

External Evaluation

University Practical examination will be for 70 marks and will test the performance, activities and creative presentations of the students with reference to the course selected:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
Total				70

VI Learning Outcomes

Upon successful completion of this course, students will be able to:

- Incorporate basic principles of aesthetics into sculpture
- Understand and apply basic concepts, styles and latest techniques of sculpting
- Explore traditional and experimental materials and design for sculpture
- Maintain a sketchbook of ideas and drawings to work out art project and to document coursework and discussions
- Understand the latest jargons, and develop collaborative skills to exhilarate the speed of accomplishing the piece of art
- Make their portfolio rich by accomplishing projects given during course
- Become familiar with varied key sculptural techniques and formal ideas through hands-on workshops and experimentation with a variety of materials and three-dimensional assignments
- Present their work with greater impact and confidence for future prospects
- Get benefitted into other subjects of their study by developing broader and all-inclusive approach to learning

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
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Semester-II

HSUS104: POTTERY AND CERAMIC ART

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				
					Theory		Practical		Total
					Internal	External	Internal	External	
II	HSUS104	Pottery and Ceramic Art	02	02	--	--	25	25	50

I. Course Objectives

- To encourage/ foster creativity among the students
- To introduce students to the craft of clay working
- To make the students able to create as well as appreciate expressive, beautiful threedimensional clay forms
- To recognize and realize the physical, emotional, and spiritual benefits of workingwith clay, and to except and come to terms with clays humbling qualities
- To build a higher-level understanding of the ceramic process, creating an awarenessof the benefits of clay as a useful tool in their art therapy studies and practices
- To develop a deeper knowledge of the ceramic process, and become more confidentwith their hand building and glazing techniques
- To help them communicate an idea or emotion through their art work

III Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Clay • Elements and Materials of Ceramic Art andShape • Prehistoric Cultures • Basics of Pinching, Slabbing and Coiling • Potter's wheel, centering the clay, forming bowlsand cylinders • Trimming and burnishing on the wheel • Sketching the Pottery Models	5

Module No.	Title/Topic	Classroom Contact Sessions
2	Types and Techniques of Making Pottery - Types of Pottery: Porcelain Pottery, Earthenware Pottery, Stoneware Pottery - Techniques of Pottery: Hand-Built Pottery: Pinch, Coil, Slab Wheel-Thrown Pottery	7
3	Methods of Making Pottery - Coil Method - Pinch Method - Slab Method	8
4	Decorating the Clays - Different Methods of Decoration - Textures in Pottery - Colors, Painting, Carving, Glazing etc. in Pottery	7
5	Contemporary Issues in Pottery and Ceramic Art - Present Day Pottery and Ceramic Art - Place of Pottery and Ceramic Art in Contemporary Art Society - Major Practitioners of Contemporary Pottery and Ceramic Art and their Contributions	3
	Total Hours	30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester

VI Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 25 marks for internal evaluation and 25 marks for external evaluation.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- ✓ Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- ✓ Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

VI. Learning Outcomes

At the end of the course, the learners will

- Have basic understanding of clay and glaze composition and formulation with emphasis on hand built ceramic forms.
- Have explored a variety of hand building methods including extended pinch, slab built and extruded forms
- Have learned firing and glazing methods for stoneware clay
- Have learned how finishing and decorating contribute or detract from the intention as an artist
- Finally, a student will also have developed a sense of appreciation regarding how a unified, coherent form that is finely crafted is beautiful in its own right.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
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Semester-II

HSUS105: MEDIA AND GRAPHIC DESIGN

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
				Contact Hours/Week	Internal	External	Internal	External	
II	HSUS105	Media and Graphic Design	02	02	--	--	25	25	50

II. Course Objectives

- To encourage/ foster creativity among the students
- To introduce students to the fundamentals of graphic designs
- To cultivate / spawn awareness among students about the significance of art and designs, art criticism and aesthetics
- To help the students understand the meanings of concept, designs, shapes, colors, print and medium
- To give the students first-hand experience of working on Graphic Software
- To develop in students an understanding of major issues, techniques and aspects of designs and print

III. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Media and Graphic Design • <i>Creating Art, Art in Context and Art as Inquiry</i> • <i>History of Graphic Design</i> • <i>Constructional, Representational, and Simplification Drawing</i>	03
2	Layout and Design • <i>Layout, Design and Aesthetics</i> • <i>Elements of Design</i> • <i>Principles of Design:</i> • <i>Harmony, Balance, Rhythm, Perspective, Emphasis, Orientation, Repetition and Proportion</i> • <i>Impact/function of Design</i> • <i>Indigenous design practices</i> • <i>Role of design in the changing social scenario</i>	07

Module No.	Title/Topic	Classroom Contact Sessions
3	Form and Space • <i>Types of Forms: Man-made, Nature</i> • <i>Types of Space: Negative and Positive</i> • <i>Composition of Form and Space to create Layout</i> • <i>Exploring Creativity</i>	06
4	Computer Graphics • <i>An Introduction to Graphic Software</i> • <i>Flash, Corel draw, Illustrator and Photoshop</i> • <i>Pre-press Process</i>	04
5	Fonts • <i>Construction of Type</i> • <i>Anatomy of Type</i> • <i>Visual Language</i> • <i>Creating Logo and Symbol</i>	04
6	Basic Print Media • <i>An Introduction to Press and its Development Phases</i> • <i>Types of Press</i> • <i>Types of Printing Technologies</i> • <i>Post-press Processes</i>	03
7	Contemporary Issues in Graphic Design • <i>Present Day Graphic Designs</i> • <i>Contemporary Designers and their Contribution</i> • <i>Major Contemporary Issues in Graphic Design</i>	03
Total Hours		30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 25 marks for internal evaluation and 25 marks for external evaluation.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

VI. Learning Outcomes

At the end of the course, a student will

- have cultivated a sense of creativity.
- be appreciative of art and designs, art criticism and aesthetics.
- be able to recognize the elements of arts in graphic design.
- have better cognizance and association with the meaning of designs, shapes, colors, print and medium.
- be able to design graphics using computer software's like Photoshop, CorelDraw, and Illustrator.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
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Semester-II

HSUS106: ART AND CRAFT

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
				Contact Hours/Week	Internal	External	Internal	External	
II	HSUS106	Art and Craft	02	02	--	--	25	25	50

II. Course Objectives

- To encourage / foster creativity among the students
- To enable the students to work through the process of bringing an idea from conception to realization
- To enable students to create artifacts that are visually expressive
- To cultivate / spawn awareness among students about the significance of art, craft and aesthetics
- To develop students' graphic skills which may work towards the realization of ideas in the creation of 2D and 3D
- To provide opportunities to students to be conversant with the use of a variety of materials, media, tools and equipment for Art and Craft

III. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	An Introduction to Art and Craft • Basic Design and Forms • Space and Geometry • Elements of Nature and Object	03
2	Paper Cutting • Study of Designs • Context of Space and Form • Types of Textures and Papers • Principles of Paper Cutting	05
3	Design from Nature • Nature as a Source of Design • Principles of Designing Natural Object • Decorative Forms • Cutting, Collaging, Embossing and Itching	08

Module No.	Title/Topic	Classroom Contact Sessions
4	Card Board Modeling • <i>Principles of Form and Space</i> • <i>Dimensions of Space and Shape</i> • <i>Process of Modeling and Decoration</i>	06
5	Print Making • <i>An Introduction to Print Making</i> • <i>A Brief History of Print Making</i> • <i>Types of Print Making</i> • <i>Processing of Print Making</i> • <i>Sketching and Drawing</i>	05
6	Contemporary Issues in Art and Craft • <i>Present-day Art and Craft</i> • <i>Using the Waste for making the Best</i> • <i>Contemporary Issues in Art and Craft</i>	03
	Total Hours	30

IV. Instruction Method and Pedagogy

Teaching will be practical - based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 25 marks for internal evaluation and 25 marks for external evaluation.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

VI. Learning Outcomes

At the end of the course, students will be

- aware about the significance of art, craft and aesthetics.
- able to create artifacts that are visually expressive.
- able to lead the ideas from conceptualization with reference to the 2D and 3D model making.
- conversant with the use of a variety of materials, media, tools and equipment for creative Art and Craft.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES
Semester-II

HSUS109: DRAMATICS

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
					Theory		Practical		Total
					Contact Hours/Week	Internal	External	Internal	External
II	HSUS109	Dramatics	02	02	--	--	25	25	50

II. Course Objectives:

- To acquaint students with the concept of performing arts
- To teach professional acting skills to the students
- To offer training in the key areas of performance
- To acquaint students with the history, theory, and aesthetic value of theatre
- To provide students extensive training in acting skills (monologues, dialogues, and group scenes from a variety of sources), dance, voice, theatre production, and rehearsal techniques, which culminates in a performance before a live audience

III. Course Outline

Module	TITLE / TOPIC	Classroom Contact Sessions
1	Introduction to Drama • Introduction to performing arts • Drama - An art, a socializing activity, & a way of learning • Form of Drama • Elements of Drama • Types of Drama	06
2	History of Drama and Contemporary Theatre • Important world dramatists & drama—from Greek to modern • Evolution of contemporary theatre in the context of developments in Indian theatre • Major Movements in Drama	06
3	Theatre Design and Techniques • Theatre Architecture • Stage craft: Set, light, costume, make up, sound, props • Theatre techniques: from selection of script to final performance	06
4	Technicalities of Stage Performance • Selection of plot and character • Improvisation • Movement	08

	• Voice, Speech, Imagination • Character Development • Scene Enactment	
5	Contemporary Trends in Drama • New Tendencies in theatre • Drama and Society • Using drama for Social Change and Education	04
	Total Hours	30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organised during the semester.

V. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks

VI. Major Learning Outcomes

At the end of the course, a student will

1. be aware about the concept of performing art and its nuances.
2. display a working knowledge of historic of drama, its development and current trends in dramatics.
3. demonstrate skills in the technical/design preparation and execution of a theatre performance.
4. demonstrate the ability to work collaboratively.
5. develop essential transferable skills in various relevant areas of the theatre.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF HUMANITIES (FOH)
Semester – II

HSUS110: CONTEMPORARY DANCE

I. Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme Contact Hours/Week	Evaluation Scheme				
					Theory		Practical		Total
					Internal	External	Internal	External	
II	HSUS110	Contemporary Dance	02	02	--	--	25	25	50

II. Course Objectives

- To introduce students to the concept of performing arts
- To develop in students, the ability to express through the form of dance
- To foster creativity and innovativeness in students
- To enhance the aesthetic sensitivity among the students
- To inculcate in student's contemporary dance techniques, philosophies, approaches, improvisation and performance disciplines
- To help students improve concentration, mental alertness, quick reflex action, physical agility and stress relief capacities
- To provide the students with complete awareness of one's own body
- To guide the students express themselves a natural way human feelings and expressions by creating harmony

III. Course Outline

Module No.	Title / Topic	Classroom Contact Sessions
1.	Introduction to dance • Dance as a Performing Art • Dance as a Medium of Expression • History and Development of Dance	4
2.	Types of Dance • Western dance and classical dance • Salsa, rumba, hip hop, tap dance, belly dance, etc. • Indian Classical Dance forms: Odissi, Bharatanatyam, Kathak, Kathakali, Kuchipudi etc. • Other Regional dance forms in India	6
3.	Basic Elements of Dance • Movements of different parts of a body for Expression • Concepts of: Nritya, Laya and Taal	4

4.	Technical Skills in Professional Contemporary Dance • Dance technique: alignment, balance, co-ordination, flexibility and control • Expressive / presentation skills: Dynamic energy, physical engagement with the given material and stage, etc. • Skills and processes of rehearsal and production: physical energy, stamina and athleticism • Musicality: clarity of timing and phrasing	6
5.	Contemporary Trends in Dance: • Prevalent trends and techniques in contemporary dance • Future trends in contemporary dance form • On Stage Performance	10
	Total hours	30

IV. Instruction Method and Pedagogy

Teaching will be practical based on the hands on experiences, live and interactive sessions. It will run in the workshop mode. Four Workshops (each of a day) will be organized during the semester.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks

V. Major Learning Outcomes

At the end of the course, a student will

- be able to develop ability to express through the form of dance.
- have enhanced aesthetic sensitivity.
- have improved concentration, mental alertness, quick reflex action, and physical agility.
- be able to express a natural way human feelings and expressions by creating harmony.
- be able to deliver contemporary dance performance.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester – II

HSUS111: Music (Vocal)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course

Objectives of the Course:

To facilitate the learners:

- To explore and understand Vocal Music
- with an opportunity to pursue the hobby of music on the campus
- to reduce stress and get energized to study better
- to explore and understanding about the Indian Music (Vocal), and its basic terms

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Vocal Music	05
2.	Elements of Indian Vocal Music	08
3.	Introduction to Ragas and Taals	08
4.	A study of forms of Indian Vocal Music	05
5.	Contemporary Trends in Indian Vocal Music	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Vocal Music	05 Hours	18 %
	Definition of Music; Concept of Vocal Music; Basic history and evolution of Indian music; Understanding the basic concepts of Indian music, including Swara, Raga, and Tala		
2.	Elements of Indian Vocal Music	07 Hours	26%
	Sound (Dhwani) and its origin; Andolan and its types; Naad and its quality; Vibration and Frequency; Sahayak Naad, 10 Thhats of Bhatkhande Ji, Saptak, Poorvang, Uttarang, Varna, Alankar, Vadi, Samvadi, Anuvadi, Vivadi, Gamak, Meend, knowledge of Taanpoora and its parts.		
3.	Introduction to Ragas and Taals	07 Hours	26%
	Learning basic ragas like Yaman and Bhupali; Understanding aroha (ascent) and avaroha (descent) of		

	ragas; Introduction to the concept of tala (rhythm); Learning common talas like Teentaal and Ektaal; Basic tala exercises and rhythmic patterns.		
4.	A study of forms of Indian Vocal Music	05 Hours	16%
	Various forms of Indian Vocal Music: Carnatic Music, Thumari, Drupad etc.; Study of Western Music and Style of Indian Music; Study of Regional Folk Music; Study of Gharana, Ravindra Sangeet and Carnatic Music		
5.	Contemporary Trends in Indian Vocal Music	04 Hours	14%
	Fusion Music; Indian Folk and Fusion; Sufi Revival; Cultural Fusion; Indian Vocal Music Festivals, Famous Indian Vocal Artists		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the concept and elements of Vocal Music
CO2	develop a clear and controlled singing voice with proper breath control and posture.
CO3	acquire the ability to sing basic scales, patterns, and simple compositions with accuracy and clarity.
CO4	understand and perform basic ragas and talas, including common rhythmic patterns.
CO5	appreciate and celebrate rich heritage of Indian Vocal Music

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

Recommended study Material:**Reference Books:**

1. "Ragas in Indian Music: A Complete Reference Source for Carnatic Ragas" by Dr. Rajan P. Parrikar - A comprehensive guide to Carnatic music ragas with historical context and audio samples.
2. "Ragas of Indian Music: Their Structure and Evolution" by Nazir Ali Jairazbhoy - An exploration of the structure and evolution of ragas in Hindustani music.
3. "The Raga Guide: A Survey of 74 Hindustani Ragas" by Joep Bor and others - A reference book with CDs that provides in-depth information on various Hindustani ragas, including their characteristics and performances.
4. "Indian Music and the West" by Gerry Farrell - A study of the impact of Indian music on Western music and culture.
5. "The Oxford Illustrated Companion to South Indian Classical Music" by Ludwig Pesch - A comprehensive resource on South Indian classical music, including Carnatic vocal music.

Web Sources:

1. https://onlinecourses.swayam2.ac.in/cec20_aso4/preview
2. <https://www.sharda.org/>
3. https://onlinecourses.swayam2.ac.in/cec22_aso2/preview
4. <https://www.udemy.com/course/learn-indian-voice-music-from-the-scratch-to-advance/>

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES (FoH)

HS212.01: Music (Instrumental) – Tabla

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	25/25	--	50	

Pre-requisite courses:

- No pre-requisites for this course

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to reduce stress and get energized to study better
- to explore and understanding about the instrument – Tabla, and its basic terms
- to play basic tals on Tabla and learn improvisation

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
6.	Introduction to Music (Tabla)	04
7.	Basic Terms	06
8.	Main Varnas of Playing Tabla	08
9.	Basic Taals	08
10.	Improvisation in Tartaal	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Music (Tabla)	04 Hours	13.33%
	• Explanation of the various parts of Tabla • Vilambit, Madhya and Dhrut Laya • Different Bols/Syllables played on Tabla • Introduction of the procedure of playing Tabla		
2.	Basic Terms	06 Hours	20%
	• Matra, Tal, Sam, Tali, Khali, Vibhag, Dugun, Avartan • Sangeet, Nad, Swar, Laya, Bol, Kisme, Kayda, Mukhda, Thai, Tigan, Chaugun, Tukda		
3.	Main Varnas of Playing Tabla	08 Hours	26.66%
	• Dhet, Gadi, Gana Kraan, KaRaana, Ghraan, GhaRaana, DhumakiTa, DhirDhir, TeTedheTe		

4.	Basic Taals	08 Hours	26.66%
	• Trital, • Jhaptal, • Dadra • Kaherva • Rupak/Tewra • Ektal/Adital		
5.	Improvisation in Tritaal	04 Hours	13.33%
	• Tita and Tirkita kayda		

Evaluation Scheme:

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester-II

HSUS113: Music (Instrumental) – Guitar

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	50	--	50	

Pre-requisite:

- Correct posture for holding the guitar and plectrum

Objectives of the Course:

Students will be able to:

- Identify the different parts of a guitar song, such as the melody, harmony, and rhythm.
- Play a variety of strumming patterns and techniques.
- Understand the different types of scales and chords.
- Analyse arpeggios and riffs in different keys.
- Learn and perform contemporary guitar techniques and blues solos using a variety of scales and chords.
- Comprehend play guitar in a variety of styles, including acoustic, rock, blues, and country.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
11.	Introduction to Structure, Tunes & Strumming	06
12.	Relation of Scales & Chords	06
13.	Arpeggios and Introduction to Riffs	07
14.	Contemporary Techniques and Blues Soloing	05
15.	Acoustic to Advance Style	06
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Structure, Tunes & Strumming	06 hours	20%
	• Basic chord progressions and strumming, Counting the beats along with strumming, Playing along with the metronome. • Introduction to the notes on the guitar and tuning procedures, Different strumming patters and how you can create your own strumming patterns.		

2.	Relation of Scales & Chords	06 hours	20%
	- Sequencing patterns (Minor & Major), 3 notes per string patterns, practicing scales using backing tracks, Formation of minor scales, Minor scale relation with major scale and visualization of the shapes. - Harmonized major scale, Chord formation theory, Introduction to CAGED system., CAGED Barre chord shapes, Extended chord formation, Some basic shapes of extended chords.		
3.	Arpeggios and Introduction to Riffs	07 hours	22%
	- Arpeggios shapes using CAGED system, Arpeggio exercises with sweep picking. - Power chords, Power chord rhythm study using down strokes and accents, Palm Muting, Popular riffs.		
4.	Contemporary techniques and Blues Soloing	05 hours	18%
	- Slides, Hammer-on, Pull-off, Vibrato, Legato exercises, introduction to Bending - How 12 bar blues chord progressions are formed, Types of Blues chord progressions, Swing rhythm in 12 bar blues, Basic Blues riffs. - Pentatonic scale shapes, Pentatonic scale sequencing Patterns, How to use pentatonic scale in blues soloing.		
5.	Acoustic to Advance Style	06 hours	20%
	- Basic finger picking exercises, Etudes and popular song riffs, Fingerstyle rhythm. - Detailed study of modes and improvisation using modes, Extended arpeggios and application, more extended chords (used in Jazz) - Advanced blues soloing, Funk rhythm study, Advanced finger style grooves, Metal riffs and scale sequences.		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	play a variety of tunes on the guitar, using different strumming patterns and techniques
CO2	understand and apply the relationship between scales and chords
CO3	play arpeggios and riffs on the guitar
CO4	learn and perform contemporary guitar techniques and blues solos
CO5	play guitar in a variety of styles, from acoustic to advanced
CO6	develop their own unique guitar sound and style

Recommended Study Material:❖ **Text book:**

1. Alternate Picking – Handbook by Clifford Martin
2. Guitar: For Beginners by Nicolas Crater

❖ **Web material:**

1. <https://www.youtube.com/watch?v=wAfbTvEeMmw&list=PLovStLXqoeZ1ncXap2csJ-FKgLUCyPQzY>
2. <https://www.youtube.com/watch?v=rTMN8rCBWkw>
3. <https://www.coursera.org/learn/guitar>
4. <https://www.coursera.org/learn/guitar-scales-chord-progressions>

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester-II

HS214.01: Music (Instrumental)- Harmonium

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to reduce stress and get energized to study better
- to explore and understanding about the instrument – Harmonium, and its basic terms
- to play basic Raags on Harmonium

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
16.	Introduction to Harmonium	04
17.	Basic Terms	06
18.	Swar	08
19.	Raag	08
20.	Tal	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Harmonium	04 Hours	13.33%
	• Understanding the parts of the harmonium • How to set up and maintain the harmonium • Basic hand positioning and posture • Introduction to the keyboard layout and finger numbering		
2.	Basic Terms	06 Hours	20%
	• Swar, Raag, Tal, Alankar, Bandish, Saptak • Sthai, Aroh, Avroh, Aalap and Bolaan		
3.	Swar	08 Hours	26.66%
	• Major Scale (Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa)		

	• Shuddha, Komal and Teevra Swars • Basic Compositions of Seven Surs		
4.	Raag	08 Hours	26.66%
	• Kalyan • Bageshri • Bhairav • Bihag • Chhayanat • Puriya Dhanashri • Bhimpalas • Jaijaiwanti • Asawari • Bibhas • Miya Malhar • Multani • Darbari Kanada • Bilawal		
5.	Tal	04 Hours	13.33%
	• Trital • Ektaal • Chautaal • Dadra • Kaherva • Rupak • Tewra • Surfakta (Sul taal) • Deepachandi • Tilwada		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify the parts of Harmonium and basic positioning before playing
CO2	Use the basic terms with understanding when it is needed in Harmonium playing
CO3	Identify and play the basic Swars of Tabla playing
CO4	Play the basics of the Raags mentioned
CO5	Identify and play along with the Tals mentioned

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester-II

HSUS115: Music (Instrumental) – Flute

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02/15	--	30	02
Marks	--	50	--	50	

Objectives of the Course:

To facilitate the learners:

- To introduce students to the fundamental techniques and principles of playing the flute.
- To develop proficiency in basic flute playing skills, including tone production, finger technique, and breath control.
- To explore different musical styles and genres through flute repertoire.
- To foster an appreciation for the artistic and historical aspects of flute music.
- To encourage individual and ensemble performance experiences.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to the Flute	04
2.	Fundamentals of Sound Production	06
3.	Finger Technique and Scales	06
4.	Flute Repertoire	06
5.	Flute Styles and History	04
6.	Ensemble Playing	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to the Flute	04 Hours	14%
	• Course orientation and objectives • Overview of the flute: parts and types • Assembly and disassembly of the flute • Proper posture and hand positioning • Introduction to basic embouchure		
2.	Fundamentals of Sound Production	06 Hours	20%
	• Breath control and support • Producing a clear and focused tone • Articulation techniques: tonguing and slurring		

	• Dynamic control and expression • Tuning and intonation		
3.	Finger Technique and Scales	06 Hours	20%
	• Basic fingerings and hand coordination exercises • Major and minor scales • Arpeggios and scale patterns • Chromatic scale • Articulation exercises		
4.	Flute Repertoire	06 Hours	20%
	• Introduction to classical, folk, and popular flute repertoire • Selecting and preparing a solo piece • Interpretation and expression in music • Performance practice and stage presence		
5.	Flute Styles and History	04 Hours	14%
	• Historical overview of flute music • Major flute composers and their contributions • Exploring different styles: classical, jazz, world music, etc. • Listening and analysis of representative pieces		
6.	Ensemble Playing	04 Hours	12%
	• Introduction to flute ensembles • Collaborative rehearsal and performance • Role of the flute in chamber music • Developing ensemble communication skills		

Course Outcome (COs):

At the end of the course, the students will be

CO1	Demonstrate proper flute posture, hand position, and embouchure technique.
CO2	Produce a clear and controlled flute tone.
CO3	Execute basic fingering patterns and scales fluently.
CO4	Interpret and perform a variety of musical pieces on the flute.
CO5	Identify key historical and stylistic characteristics of flute music.
CO6	Participate in ensemble performances and collaborate effectively with other musicians.

Recommended Study Material:

❖ Reference books:

1. *The Flute Book: A Complete Guide for Students and Performers* by Nancy Toff
2. *Trevor Wye's Practice Books for the Flute* by Trevor Wye
3. *Flute Basics: A Method for Individual and Group Learning* by Sally Adams and Tracey Rush
4. *The Art of Flute Playing* by Edwin Putnik
5. *Selected Flute Solos: Everybody's Favorite Series, Volume 101*
6. *The Flute and Flute-Playing* by Theobald Boehm
7. *Flute Music by French Composers* edited by Louis Moyse

❖ Online Courses

1. Coursera - Introduction to Classical Music (Offered by Yale University)
2. Udemy - Flute Lessons for Beginners
3. Udemy - Intermediate Flute Lessons
4. edX - The Science of Flute Playing (Offered by MIT)
5. Skillshare - Flute: Start Playing Songs in 7 Lessons

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convener / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester-II

HSUS116: Indian Classical Dance – Kathak

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course.

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to develop a flair for performing arts.
- to acquaint students with Indian classical dance form, and its basic terms
- make them aware with basic mudras, abhinayas, and steps of the dance form.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Kathak	06
2.	Footwork and Tatkar	06
3.	Abhinaya (Emotive Expressions)	06
4.	Toda and Tukda (Compositions)	06
5.	Ghungroo (Ankle Bells) and Rhythmic Patterns	06
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Kathak	06 Hours	20%
	• Understand the historical and cultural significance of Kathak. • Learn the basic postures (Tal Poses) and their importance. • Introduction to hand gestures (Mudras) and their meanings. • Develop a basic understanding of rhythm (Tala) in Kathak.		

2.	Footwork and Tatkar	06 Hours	20%
	- Learn and practice basic footwork patterns (Tatkar). - Develop proper foot placement and coordination. - Understand the importance of rhythm and synchronization in Kathak. - Explore the concept of Nritya (pure dance) in Kathak.		
3.	Abhinaya (Emotive Expressions)	06 Hours	20%
	- Understand the concept of Abhinaya in Kathak. - Learn and practice facial expressions (Bhava) and emotions. - Explore storytelling through dance.		
4.	Toda and Tukda (Compositions)	06 Hours	20%
	- Learn and perform basic Kathak compositions (Toda and Tukda). - Understand the structure of Kathak compositions. - Practice and refine the choreography.		
5.	Ghungroo (Ankle Bells) and Rhythmic Patterns	06 Hours	20%
	- Explore the significance of Ghungroo (ankle bells) in Kathak. - Learn to wear and produce rhythmic sounds with Ghungroo. - Practice and execute various rhythmic patterns (Bols). - Understand the importance of laya (tempo) in Kathak.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	remember basic postures and hand gestures of Kathak.
CO2	understand the concept of Nritya in Kathak.
CO3	design storytelling through dance and will grasp the concept of Abhinaya and its role in Kathak.
CO4	perform basic Kathak Toda and Tukda compositions.
CO5	apply and create rhythmic sounds effectively with Ghungroo executing various rhythmic patterns through laya in Kathak.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF HUMANITIES (FoH)
Semester-II

HSUS117: Indian Classical Dance – Bharatanatyam

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	50	--	50	

Pre-requisite courses:

- No pre-requisites for this course.

Objectives of the Course:

To facilitate the learners:

- with an opportunity to pursue the hobby of dance on the campus
- to develop a flair for performing arts.
- to acquaint students with Indian classical dance form, and its basic terms
- make them aware basic mudras, abhinayas, and steps of the dance form.

Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to Bharatanatyam	06
2.	Adavus and Footwork	06
3.	Abhinaya (Emotive Expressions)	06
4.	Rasa and Bhava	06
5.	Choreography and Performance	06
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours:	30

Detailed Syllabus:

1.	Introduction to Bharatanatyam	06 Hours	20%
	• Understand the historical and cultural significance of Bharatanatyam. • Learn the basic postures (Asanas) and their importance. • Introduction to hand gestures (Mudras) and their meanings. • Develop a basic understanding of rhythm (Tala) in Bharatanatyam.		
2.	Adavus and Footwork	06 Hours	20%

	- Learn and practice basic Adavus (fundamental dance steps). - Develop proper footwork and coordination. - Understand the importance of rhythm in Bharatanatyam. - Explore the concept of Nritta (pure dance) in Bharatanatyam.		
3.	Abhinaya (Emotive Expressions)	06 Hours	20%
	- Understand the concept of Abhinaya in Bharatanatyam. - Learn and practice facial expressions (Bhava) and emotions. - Explore storytelling through dance.		
4.	Rasa and Bhava	06 Hours	20%
	- Explore the nine Rasas (emotions) in Bharatanatyam. - Learn how to evoke different emotions through expressions and movements. - Understand the importance of emotional connectivity in dance.		
5.	Choreography and Performance	06 Hours	20%
	- Learn a simple Bharatanatyam composition. - Understand the structure of a dance performance. - Practice and refine the choreography. - Gain confidence in performing in front of an audience.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	understand the historical and cultural significance of Bharatanatyam.
CO2	analyse proper footwork and coordination.
CO3	design storytelling through dance.
CO4	perform to evoke different emotions through expressions and movements.
CO5	remember the choreography and gain confidence in performing in front of an audience.

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

- Formative: Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks
- Summative: External / Semester End Evaluation (SEE) – 25 Marks

Internal / Continuous and Comprehensive Evaluation (CCE) – 25 Marks	
Exam Pattern	Marks
Lab Work Assessment (Best 3 out of 4)	45
Viva voce/ Lab Quiz (Best 3 out of 4) (Note: Quiz Number 1 and 3 will be administered as lab quizzes, while Quiz Number 2 and/or 4 shall serve as part of the mid-term evaluation.)	45
Attendance	10
Total	100* (scaled to 25 marks)

*Note: The total Internal / CCE score out of 100 marks will be converted to 25 marks.

External / Semester End Evaluation (SEE) – 25 Marks	
Exam Pattern	Marks
Quiz	40
Viva-voce	30
Presentation/Task/GD/Case Analysis etc. (The Course Convenor / Faculty will brief the students about the examination components and weightage)	30
Total	100* (scaled to 25 marks)

*Note: The total External / SEE score out of 100 marks will be converted to 25 marks.